01st Sep 2026

재생에너지 기술 및 응용

(Renewable Energy Technologies and Applications)

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Hydrogen **E**nerg**Y** Lab **ON** **E**arth

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Course Introduction

- **Lecture: Renewable Energy Technologies and Applications (L01211101)**
  - Class time: Tuesday, 12:00–14:50

- **Professor**: Haewon Seo
  - Office: Prof Res Bldg , Rm 202
  - E–mail: heyone@hufs.ac.kr
  - Office hour: Wednesday (till 17:00)

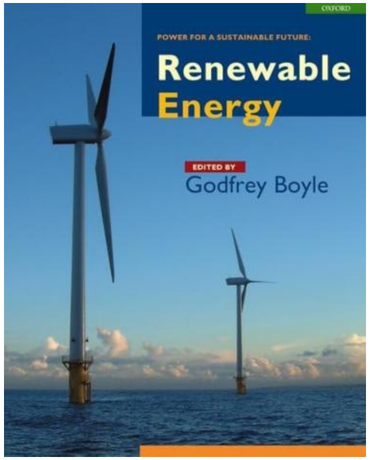
- **Scope**
  - This course explores the necessity of renewable energy in meeting the surging power demands of the AI era. Grounded in thermodynamics and fluid mechanics, it examines the operating principles and technologies of major renewable systems. Furthermore, it reviews the latest technological trends and discusses system integration strategies for practical applications.

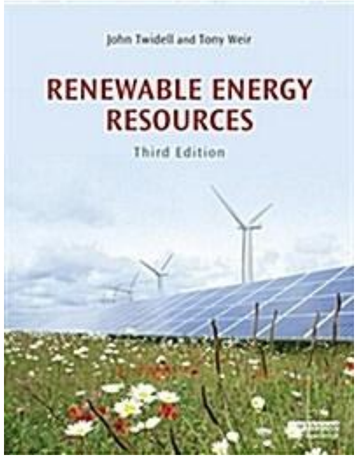
Course Introduction

❑ Textbook

– Main

- Renewable Energy (G. Boyle), 2nd Ed., Oxford University Press (2004)
- Renewable Energy Sources, (J. Twidell & T. Weir), 3rd Ed., Talyor & Francis (2015)
- Thermodynamics (Y.A. Cengel et al.), 9th Ed., McGraw-Hill (2018)





– Reference

- Scientific articles
- IPCC, Corpernicus Climate Change, NOAA, NASA, etc.

Course Introduction

❑ **Schedule**; to be flexibly operated

Week	Date	Contents
01st	**Sep 01st 2026**	**Introduction to Energy**
02nd	Sep 08th 2026	Fundamentals of Thermodynamics
03rd	Sep 15th 2026	Fossil Fuels
04th	Sep 22nd 2026	Solar Thermal Energy
05th	Sep 29th 2026	Solar Photovoltaics (1)
06th	Oct 06th 2026	Solar Photovoltaics (2)
07th	Oct 13th 2026	Presentation (1)
08th	Oct 20th 2026	Midterm Exam
09th	Oct 27th 2026	Fundamentals of Fluid Dynamics
10th	Nov 03rd 2026	Wind Energy
11th	Nov 10th 2026	Hydropower
12th	Nov 17th 2026	Bioenergy
13th	Nov 24th 2026	Waste Energy & Geothermal Energy
14th	Dec 01st 2026	Presentation (2)
15th	Dec 15th 2026	Final Exam

Course Introduction

❑ Evaluation

– Attendance (10%)

– Assignment (20%)

– Midterm Exam (30%)

– Final Exam (40%)

1. Introduction to Energy

Contents

What is Energy?

(*a*) Refrigerator

(*b*) Boats

(*c*) Aircraft and spacecraft

(*d*) Power plants

(*e*) Human body

(*f*) Cars

(*g*) Wind turbines

(*h*) Food processing

(*i*) A piping network in an industrial facility.

What is Energy?

❑ Energy

The quantitative property transferred to a system or body, recognizable as the capacity to do work, produce change, or emit heat and light

- **Energy is a scalar quantity (has magnitude, no direction)**

- **Total energy in a closed system is conserved over time**

$$\frac{dE_{\text{total}}}{dt} = 0 \Leftrightarrow E_{\text{total}} = \text{constant}$$

- **Units: J (SI), Wh, cal, eV, etc.**

1 Wh = 3600 J (Electrical energy)

1 cal = 4.184 J (Metabolic energy)

1 eV = 1.602×10^{-19} J (Particle/Subatomic energy)

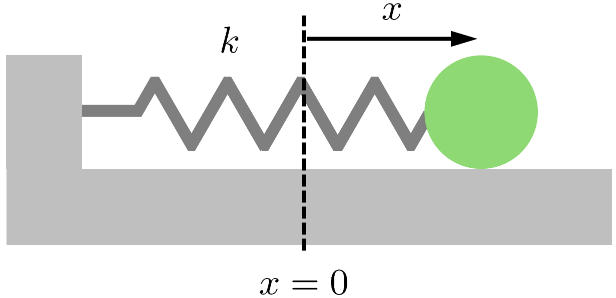
Forms of Energy

❑ Macroscopic Forms

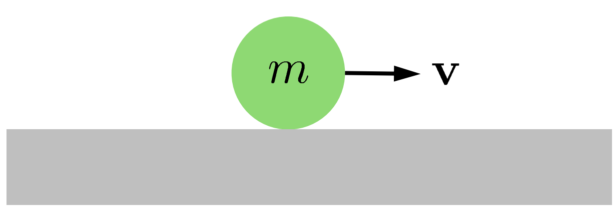
Possessed by a system as a whole relative to an external reference frame

(e.g.) Potential energy, Kinetic Energy

$$U = \frac{1}{2}kx^2$$



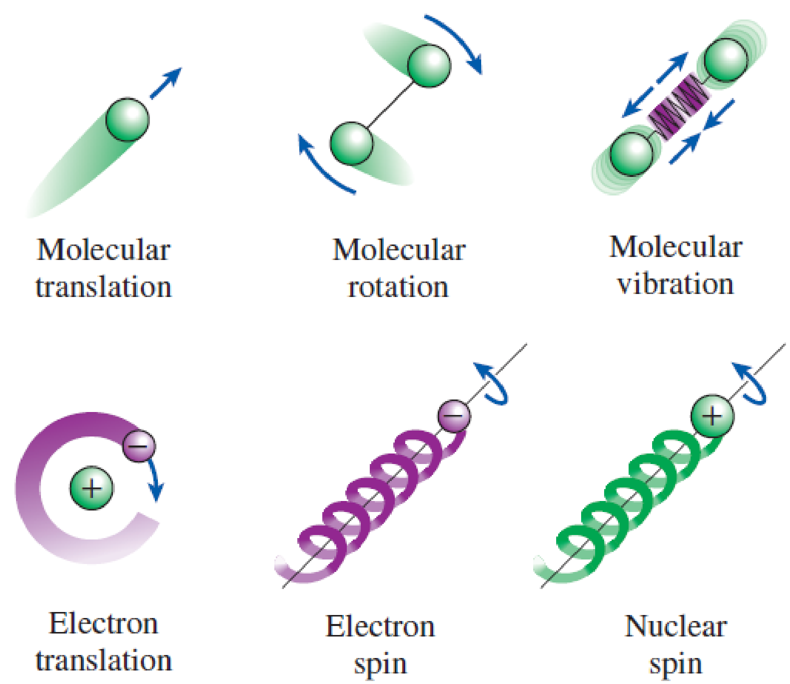
$$K = \frac{1}{2}mv^2$$



❑ Microscopic Forms

Related to the molecular structure and activity of a system, and independent of external reference frames.

The sum of all the microscopic forms of energy is called *the internal energy* of a system.



Forms of Energy

❑ Potential Energy

Stored energy that depends on the relative positions of objects

– Chemical Energy
(e.g., Gasoline, Food)

– Mechanical Energy
(e.g., Compressed spring)

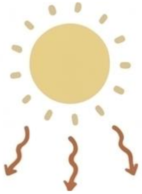
– Nuclear Energy
(e.g., Uranium fuel)

– Gravitational Energy
(e.g., Water behind a dam)

❑ Kinetic Energy

Energy that a system possesses as a result of its motion relative to some reference frame

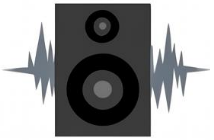
– Radiant Energy
(e.g., Sunlight, Lamp light)



– Thermal Energy
(e.g., Boiling water)

– Motion Energy
(e.g., A moving car)

– Sound Energy
(e.g., A clapping sound)



International System of Units

❑ SI units

A coherent system of units of measurement starting with seven base units, including **kg** (mass), **m** (length), **s** (time), **A** (electric current), **K** (temperature), **cd** (luminous intensity), and **mol** (amount of matter)

- **MLT dimension system**
  - M ⇔ kilogram (kg), L ⇔ meter (m), T ⇔ second (s)
  - (e.g.) [Velocity] = $L\ T^{-1}$ ⇔ $m\ s^{-1}$, [Volumetric flow] = $L^3\ T^{-1}$ ⇔ $m^3\ s^{-1}$

Variables	Unit (SI)	MLT
Acceleration	$m\ s^{-2}$	LT^{-2}
Angle	rad	$M^0L^0T^0$
Area	m^2	L^1
Density	$kg\ m^{-3}$	ML^{-3}
Energy	J	ML^2T^{-2}
Force	N	MLT^{-2}
Frequency	Hz	T^{-1}
Heat capacity	$J\ kg^{-1}\ K^{-1}$	$L^2T^{-2}\Theta^{-1}$
Length	m	L
Mass	kg	M

Variables	Unit (SI)	MLT
Power	W	ML^2T^{-3}
Pressure	Pa	$ML^{-1}T^{-2}$
Strain	–	$M^0L^0T^0$
Stress	Pa	$ML^{-1}T^{-2}$
Surface tension	$N\ m^{-1}$	MT^{-2}
Temperature	K	Θ
Time	s	T
Viscosity (dyn.)	Pa s	$ML^{-1}T^{-1}$
Viscosity (kin.)	$m^2\ s^{-1}$	L^2T^{-1}
Volume	m^3	L^3

International System of Units

❑ Metric Prefix

A unit prefix that precedes a basic unit of measure to indicate a multiple or submultiple of the unit

Multiple	Prefix	Adoption
10^{30}	quetta, Q	2022
10^{27}	ronna, R	
10^{24}	yotta, Y	1991
10^{21}	zetta, Z	
10^{18}	exa, E	1975
10^{15}	peta, P	
10^{12}	tera, T	1960
10^{9}	giga, G	
10^{6}	mega, M	1873
10^{3}	kilo, k	1795
10^{2}	hecto, h	
10^{1}	deka, da	

Multiple	Prefix	Adoption
10^{-1}	deci, d	1795
10^{-2}	centi, c	
10^{-3}	mili, m	
10^{-6}	micro, μ	1873
10^{-9}	nano, n	1960
10^{-12}	pico, p	
10^{-15}	femto, f	1964
10^{-18}	atto, a	
10^{-21}	zepto, z	1991
10^{-24}	yocto, y	
10^{-27}	ronto, r	2022
10^{-30}	quecto, q	

Force and Energy

❑ Force

Any action that tends to maintain or alter the motion of a body or to distort it

$$\mathbf{F} = \frac{d\mathbf{p}}{dt} = m\mathbf{a} \qquad \text{(Newton's 2}^{\text{nd}}\text{ Law)}$$

- **[Force] = [Mass][Acceleration] = MLT^{-2} ⇔ kg m s^{-2} = N**

  e.g., the gravitational forces on an 80.0-kg man on Earth and the Moon are respectively

  $F_{g,Earth}$ = 80.0 kg × 9.81 m s^{-1} = 785 N = 80.0 kgf

  $F_{g,Moon}$ = 80.0 kg × 1.62 m s^{-1} = 130 N = 13.1 kgf

❑ Energy

The capacity for performing work

$$W = \int \mathbf{F} \cdot d\mathbf{r}$$

- **[Energy] = [Force][Length] = ML^2T^{-2} ⇔ kg m^2 s^{-2} = J**

  e.g., the kinetic energy of an 80.0-kg man who runs at 10.0 m s^{-1} is

  K = 1/2 × 80.0 kg × (10.0 m s^{-1})^2 = 4000 J (c.f., daily adult caloric intake = ~10 MJ)

Power and Pressure

Power

The rate at which work is performed

$$P = \frac{dW}{dt}$$

- [Power] = [Energy][Time]$^{-1}$ = ML^2T^{-3} ⇔ J s^{-1} = W

  e.g., the power an 80.0-kg man exerts climbing stairs at 1.00 m s^{-1} is

  P = 80.0 kg × 9.81 m s^{-2} × 1.00 m s^{-1} = 785 W (c.f., 1 horsepower = 746 W)

Pressure

The perpendicular force per unit area, or the stress at a point within a confined fluid

$$p = \frac{F}{A}$$

- [Pressure] = [Force][Area]$^{-2}$ = $ML^{-1}T^{-2}$ ⇔ N m^{-2} = Pa

  e.g., the pressure exerted on the ground by an 80.0-kg man with a total contact area of 0.0450 m^2 is

  p = 80.0 kg × 9.81 m s^{-2} / 0.0450 m^2 = 17.4 kPa (c.f., p_{atm} = 101.325 kPa)

Power Generation

❑ Quiz (1)

A household is paying KRW 215 kWh^{-1} for electric power. To reduce its power bill, the household installed a solar panel with a rated power of 3 kW. If the solar panel operates 1400 hours per year (~4 hours per day) at the rated power, determine *the amount of electric power generated by the solar panel* and *the money saved by the household per year*.

Total energy = Energy per unit time × Time interval
=
= **kWh yr^{-1}**

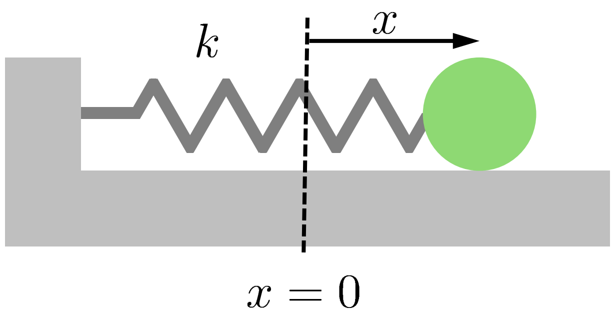
Money saved = Total energy × Unit cost of energy
=
= **KRW yr^{-1}**

Energy Conservation

❑ Quiz (2)

Consider a one-dimensional motion along the *x*-axis of a spring–mass system, where a ball of mass *m* is attached to a spring with a spring constant *k*. Assume that no nonconservative forces are acting on the system. Solve the following problems:

*friction, air resistance



- Write down *the equation of motion* for the system.

- Determine *the angular frequency ω of the oscillation* using *k* and *m*.

Energy Conservation

❑ Quiz (2) (Cont'd)

Consider a one-dimensional motion along the *x*-axis of a spring–mass system, where a ball of mass *m* is attached to a spring with a spring constant *k*. Assume that no nonconservative forces are acting on the system. Solve the following problems:

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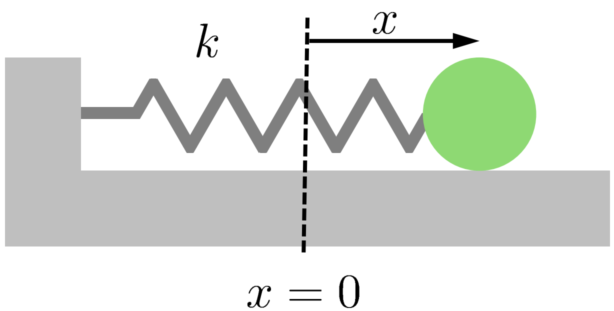


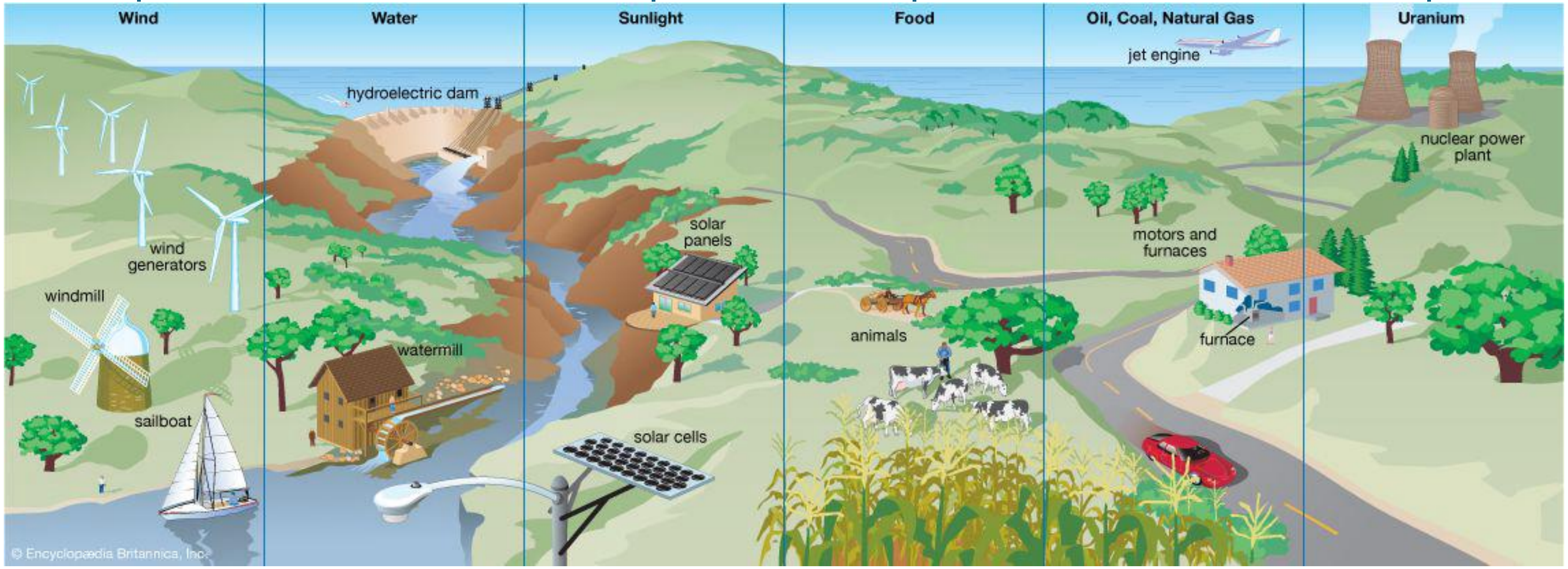
- Derive *the displacement x(t)* using the amplitude x_0, *k*, and *m.*

- Estimate *the total energy of the system* E_{total} and prove that it is constant over time.

Energy in Daily

❑ Energy Resources

<Inexhaustible energy> <Exhaustible energy>



– Harnessing the power of *nature* (e.g., wind, water, and sunlight)

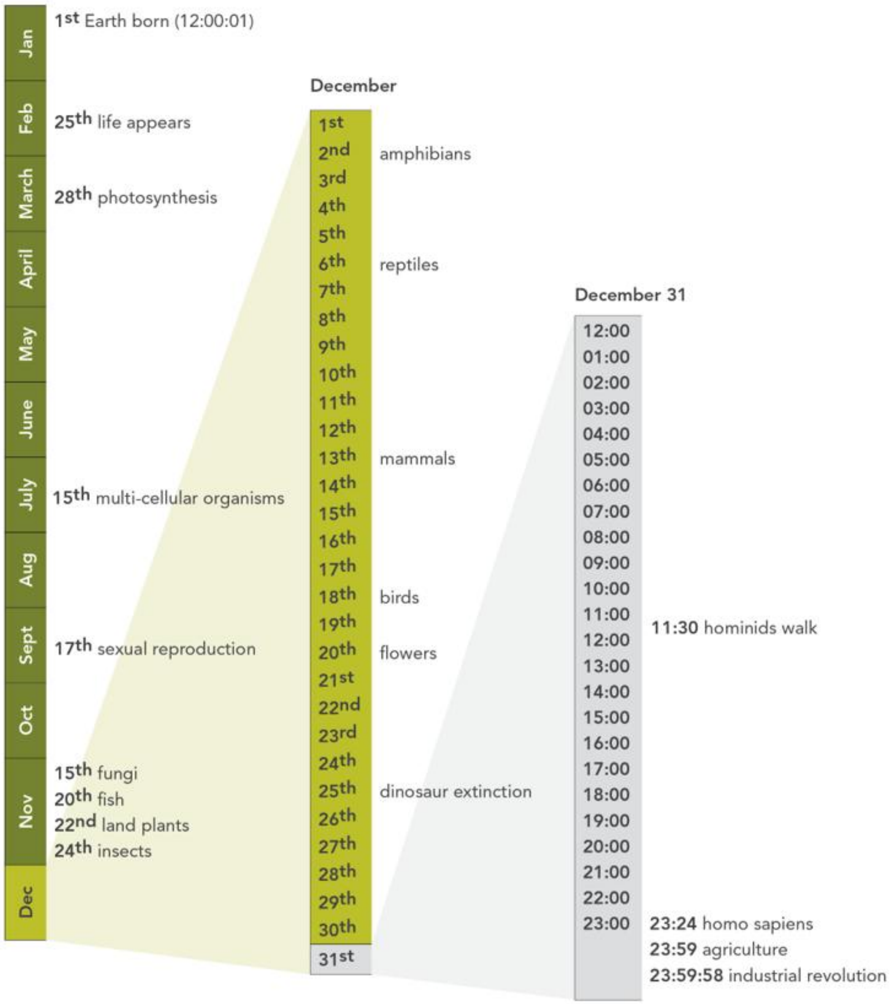
– Foods provided by *plants and animals*

– Burning *fossil fuels* (e.g., oil, coal, and natural gas) for energy

– Nuclear energy from *radioactive atoms* (e.g., uranium, thorium, etc.)

Timeline of Energy

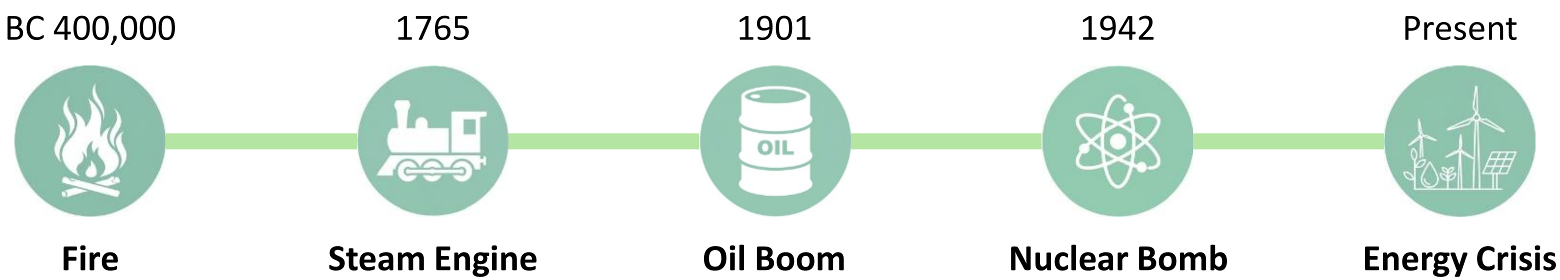
❑ Geologic Calendar



<Source: Biomimicry 3.8>

Timeline of Energy

❑ Energy Transition



Sir Benjamin Thompson (1753–1814)	*The Nature of Heat (1798)* “Heat is a form of energy corresponding to a definite amount of mechanical work.”
Julius Robert von Mayer (1814–1878)	*The Laws of Conservation of Energy (1842)* “Energy can be neither created nor destroyed”
William John Macquorn Rankine (1820–1872)	*Potential Energy (1853)*
William Thomson, Lord Kelvin (1824–1907)	*Kinetic Energy (1867)*
Albert Einstein (1879–1955)	*The Theory of Relativity (1905)* “Mass and energy are equivalent and interchangeable”

Types of Energy

❑ Nonrenewable Energy

Energy derived from natural resources that cannot be readily replenished in a short period

*Millions of years to be formed

Oil

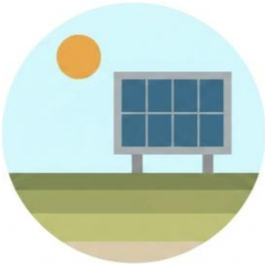
Coal

Natural Gas

Nuclear

❑ Renewable Energy

Energy derived from natural resources that are replenished on a human timescale



Solar

Wind

Hydropower

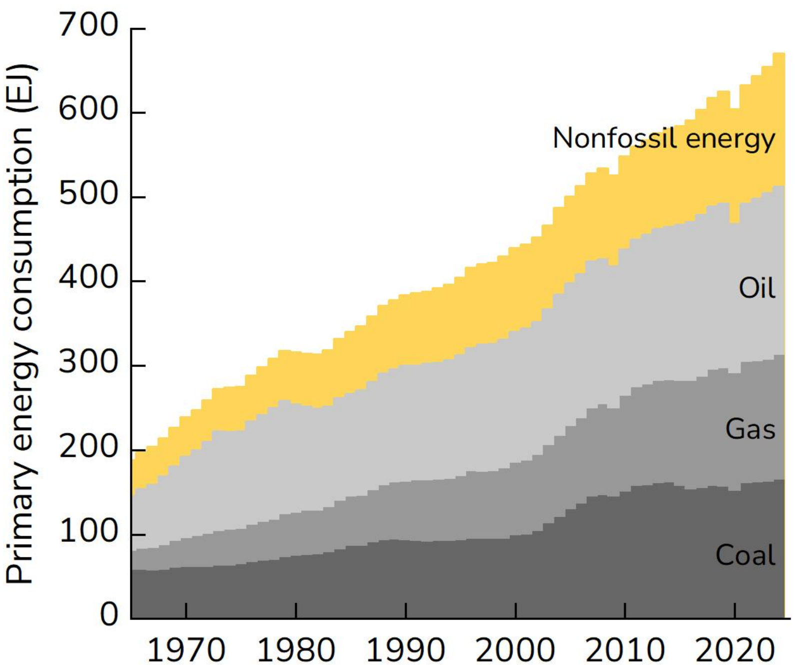
Biomass

Geothermal Heat

Types of Energy

❑ Global Energy Consumption

The world relies on fossil fuels, which account for ~80% of primary energy consumption



<Source: Statistical Review of World Energy (2025)>

	Coal	Natural Gas	Oil
Energy Use (2024)	165 EJ (~25%)	149 EJ (~22%)	199 EJ (~30%)
Left (2020)	139 yr	49 yr	56 yr
Efficiency (2024)	33%	44%	31%
GHG emission (2024)	15.8×10^9 t (~41%)	8.01×10^9 t (~21%)	12.5×10^9 t (~32%)

<Source: Form EIA-923 (2026), Global Carbon Budget (2025)>

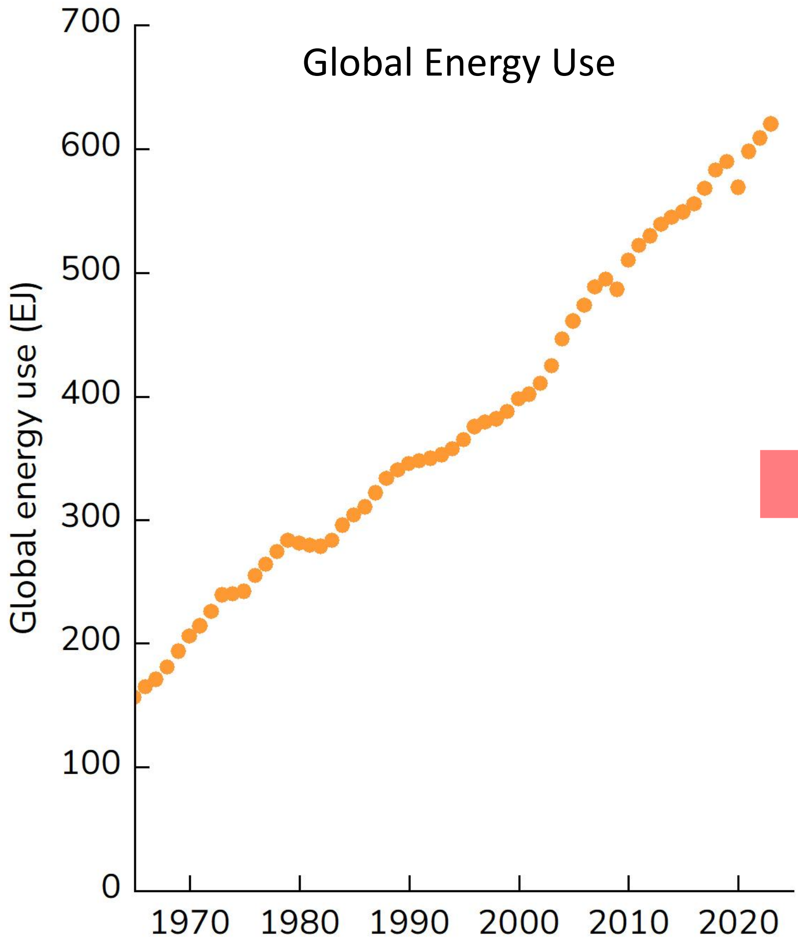
– Political issues: Foreign sources, Resource wars, Carbon taxes

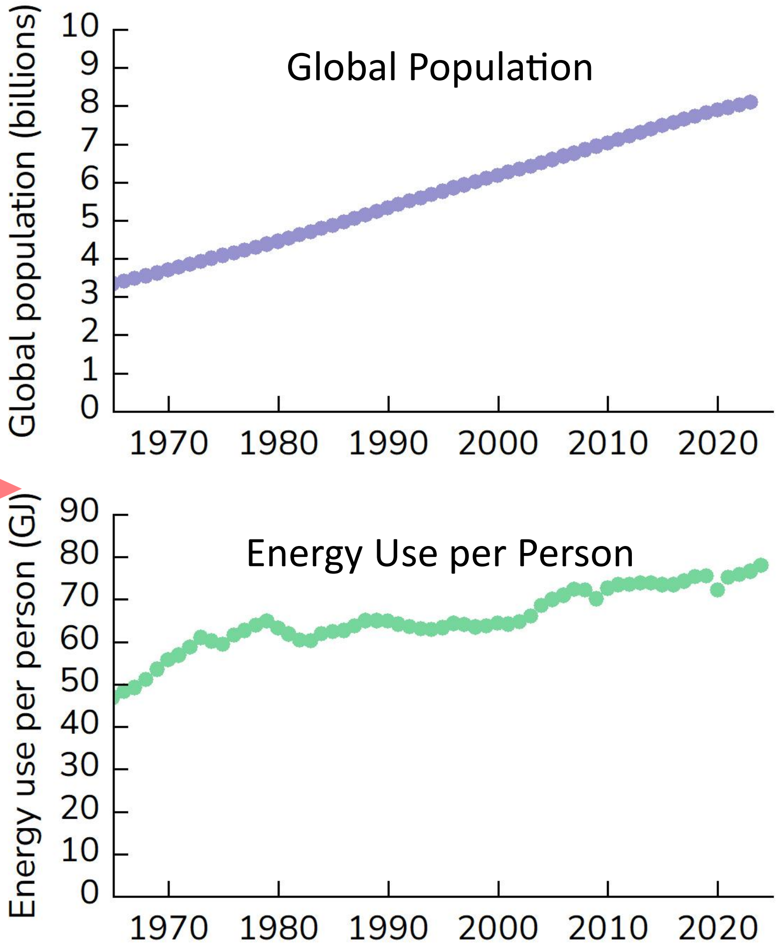
– Economical issues: Increased global demand, Limited resource

– Environmental issues: Climate crisis due to CO_2, NO_x, smog, etc.

Types of Energy

❑ Global Energy Use





– More and more energy will be needed in near future

Types of Energy

Transformation

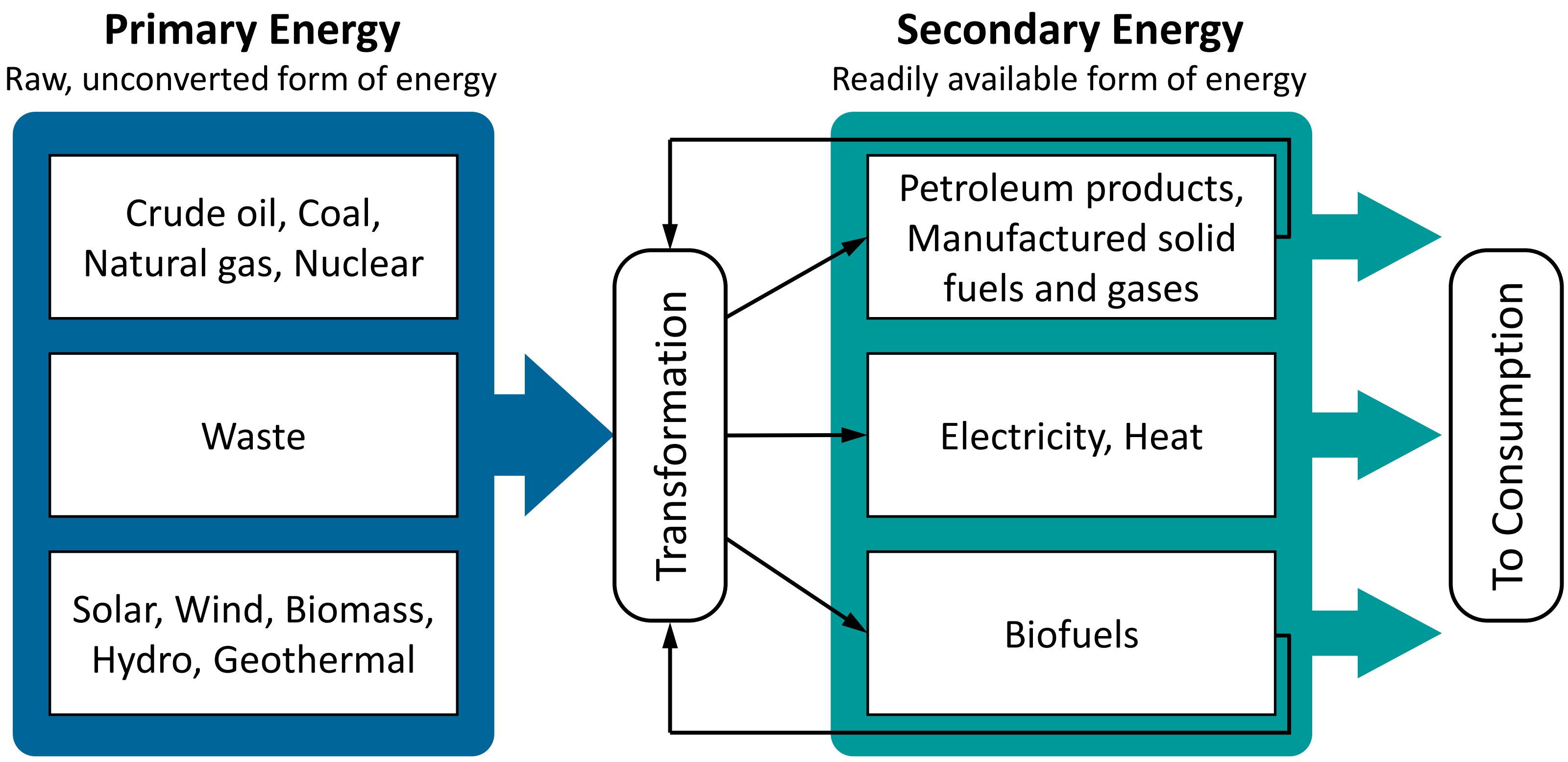
Transformation

– Transforming a raw, unconverted form into a useful form is necessary

Types of Energy

❑ Primary and Secondary Energy

Energy is indeed abundant, but *usable* energy is strictly limited, making *tools* essential



– *Usable* energy must be transformed from primary energy sources

– Every energy conversion process must entail *losses*

Types of Energy

❑ Examples of Transformation

System	From	To
ATP Hydrolysis	Chemical	Mechanical
Battery	Chemical	Electrical
Electric Generator	Kinetic Mechanical	Electrical
Electric Heater	Electrical	Thermal
Fire	Chemical	Thermal Light
Friction	Kinetic	Thermal
Fossil Fuel	Chemical	Thermal Electrical Mechanical
Fuel Cell	Chemical	Electrical
Geothermal Power	Thermal	Electrical

System	From	To
Heat Engine	Thermal	Mechanical
Hydroelectric Dam	Gravitational	Electrical
Electric Lamp	Electrical	Thermal Light
Microphone	Sound	Electrical
Ocean Thermal Power	Thermal	Electrical
Photosynthesis	Radiation	Chemical
Piezoelectrics	Strain	Electrical
Thermoelectric	Thermal	Electrical
Wave Power	Mechanical	Electrical
Windmill	Wind	Electrical Mechanical

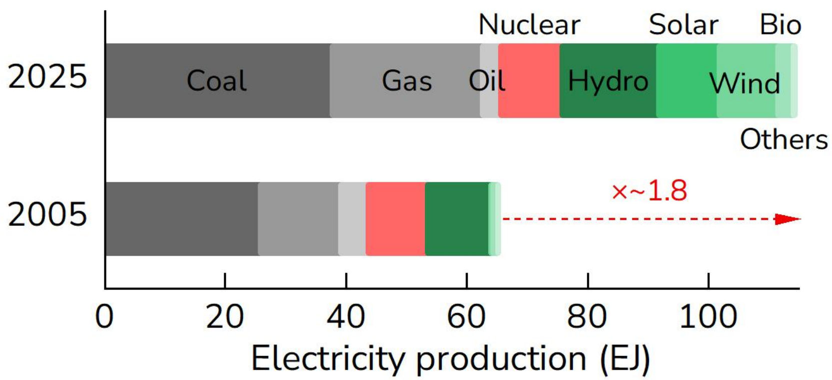
– There are many different systems, both natural and engineered

Overview of Global Electricity

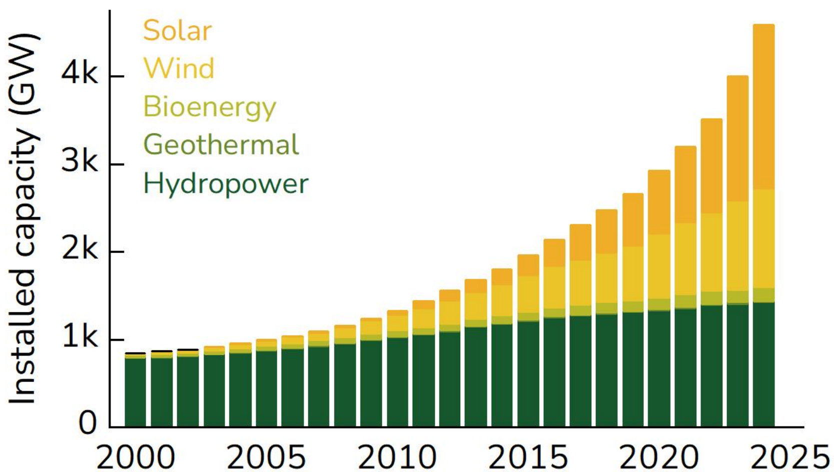
❑ Global Electricity Production

RE accounts for 34% in 2025 (18% in 2005)

*Others: geothermal, biomass, waste, wave and tidal

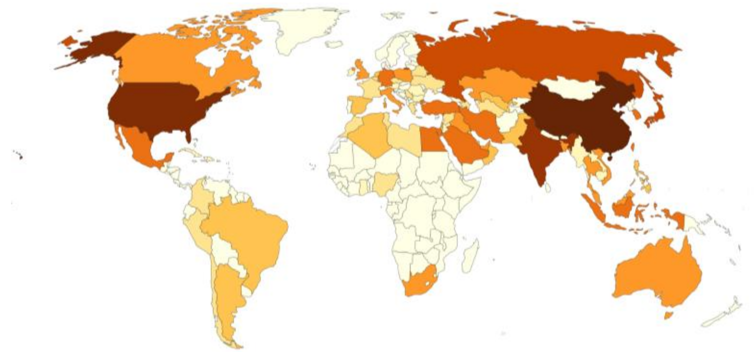


<Source: Statistical Review of World Energy (2025)>

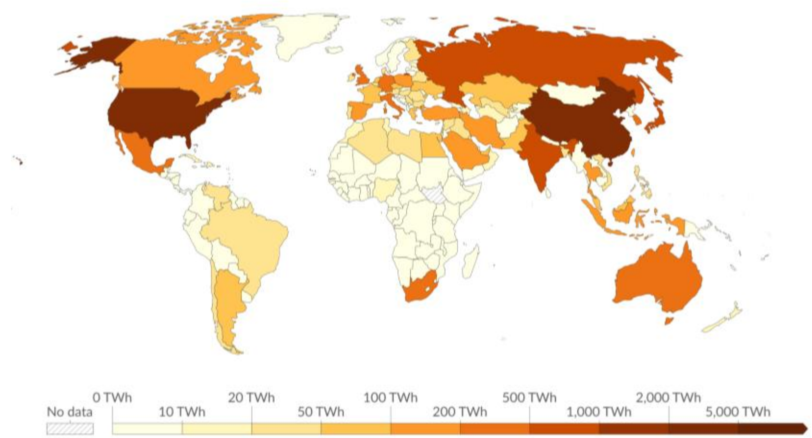


<Source: IRENA (2025)>

Electricity from Fossil Fuels (2025)



Electricity from Fossil Fuels (2005)



<Source: Our World in Data>

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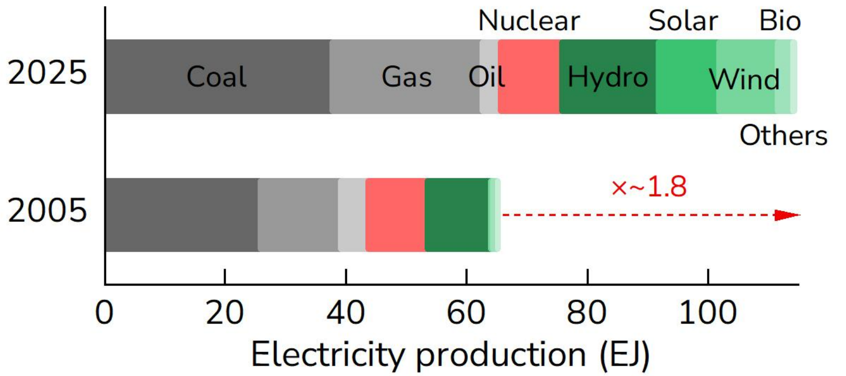

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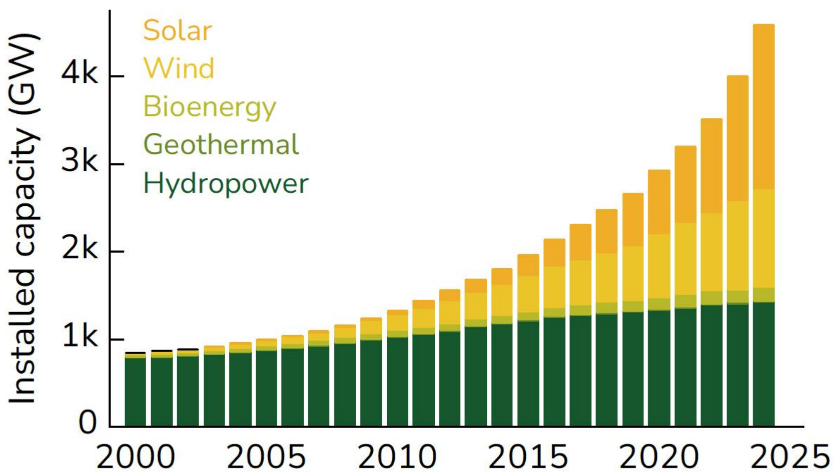
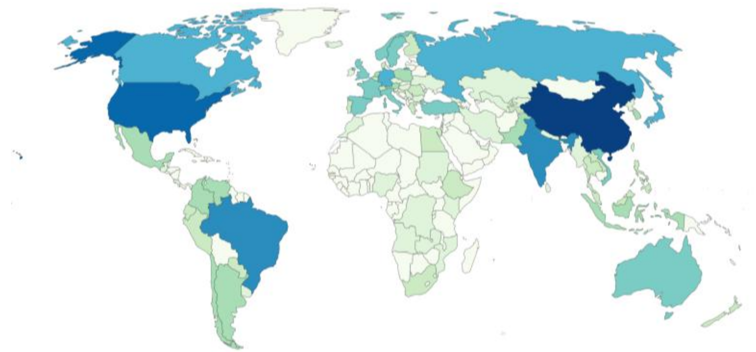
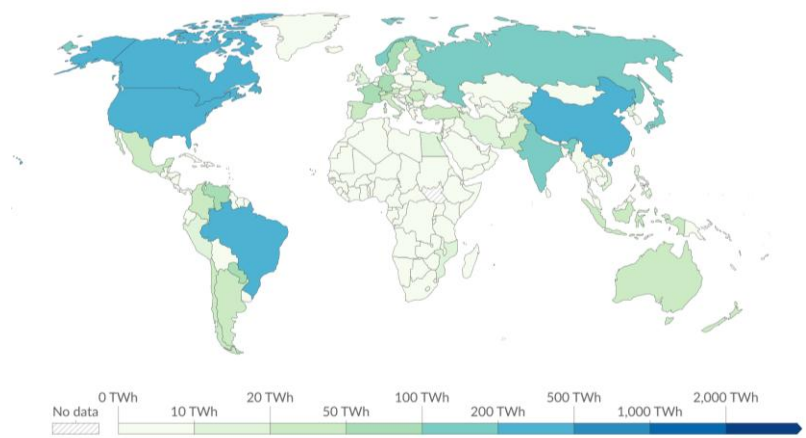


<Source: IRENA (2025)>

Electricity from Renewable (2025)



Electricity from Renewable (2005)



<Source: Our World in Data>

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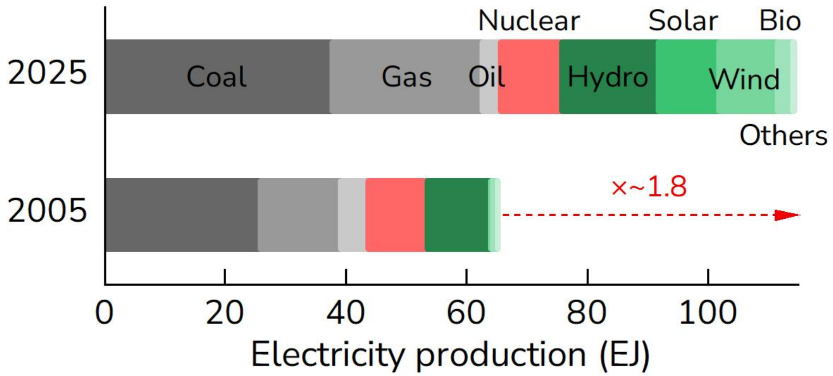

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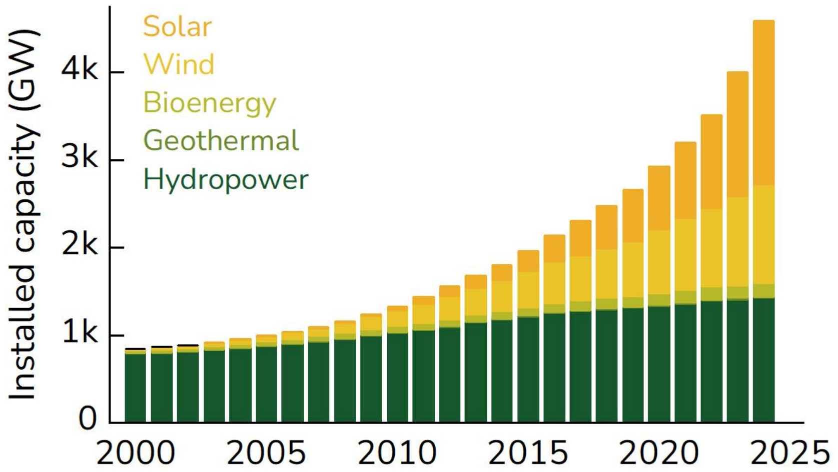
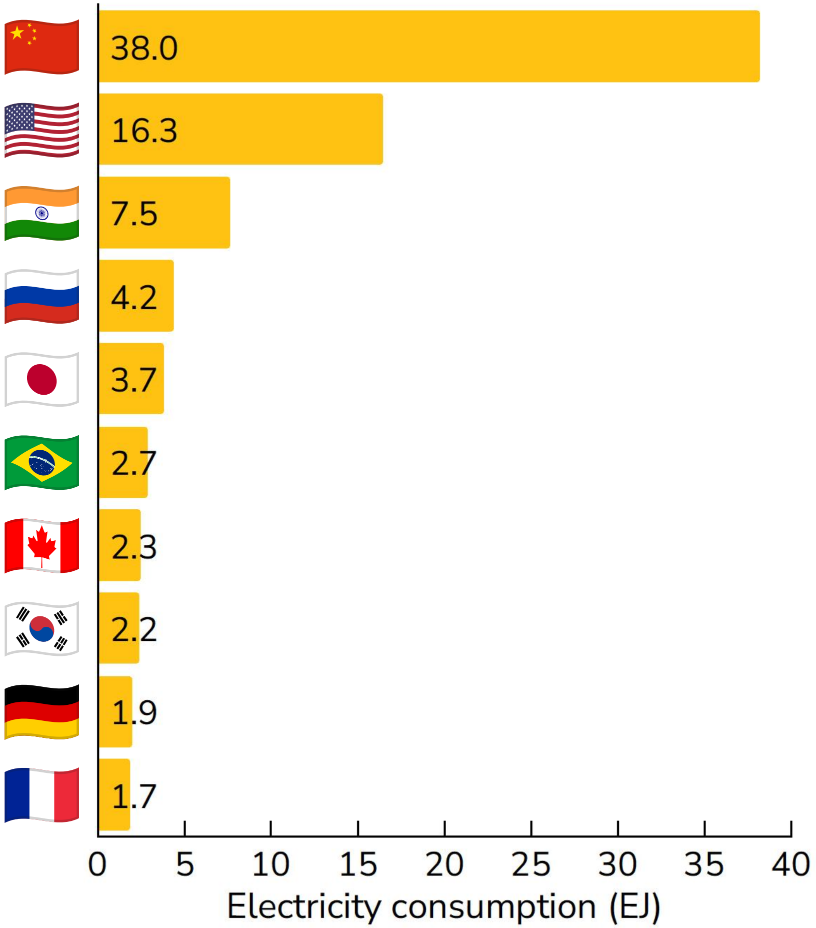


<Source: IRENA (2025)>

❑ Global Electricity Demand

Data centers consumed ~1.8 EJ in 2025

**Usage of ~3.4 EJ in 2030



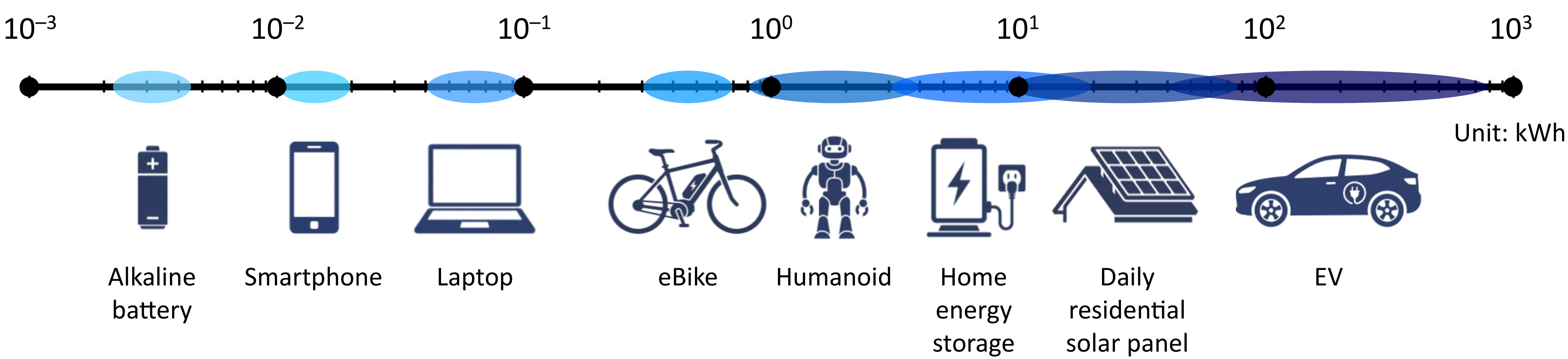
<Source: Ember (2026), Energy and AI (2026)>

Scale for Electrical Energy

❑ What can 1 kWh (= 3.6×10^6 J = 3.6 MJ) do?

1 kWh = 1000 Wh; 1 kW (= 1000 W) of power can be used for an hour

- (EV) Drive a Tesla Model Y for ~6 km
- (TV) View an LG StanbyME 2 for ~43 h
- (Laptop) Power ~14 14-inch Macbook Pros
- (Smartphone) Fully charge ~70 iPhone 17s
- (Light Bulb) Operate a 12 W light bulb for ~83 h



Scale for Electrical Energy

❑ What can 1 TWh (= 3.6×10^{15} J = 3.6 PJ) do?

1 TWh = 1,000,000,000,000 Wh = 1,000,000,000 kWh = 1,000,000 MWh = 1,000 GWh

- (EV) Drive around the world ~150,000 times in a Tesla Model Y
- (EV) Fully charge ~17,000,000 Tesla Model Ys
- (Home) Power ~300,000 Korean households for a year
- (Industry) Run Samsung for 2 weeks (Annual consumption: ~26 TWh))
- (City) Power the entire city of Seoul for a week (Annual consumption: ~50 TWh)

Alternative Energy Sources

Solar

- Solar Thermal Energy
  (Thermal process)
- Photovoltaics (PV)
  (Quantum process)

☺
- Ultimate source of most energy
- Vast potential

☹
- Low efficiency
- Limited upscaling, distribution

Nuclear

- Fission, Fusion
  (Quantum process)

☺
- High power output
- Vast potential

☹
- Toxic waste
- Disaster risk

Others

- Wind, Hydro, Wave
  (Mechanical process)
- Ocean, Geothermal
  (Thermal process)

☺
- Clean energy sources
- Substantial potential

☹
- Intermittency
- Geographic limitation

Alternative Energy Sources

❑ Drawbacks of Renewables

Compared to nuclear power, renewables offer lower LCOE but exhibit lower capacity factors

- **Levelised Cost of Electricity, *LCOE***

  LCOE = discounted sum of costs over lifetime / discounted sum of electricity over lifetime

- **Capacity Factor, *CF***

  $CF = E_{total} / (P_n \times t)$

  (E_{total} : total electricity over the time period t

  P_n: installed capacity of the device or system)

(e.g.) Three Gorges Dam (China) is, with its installed capacity of 22,500 MW (2017), the largest power generating station in the world by installed capacity. In 2015 it generated 87 TWh, for a capacity factor of **87 TWh yr^{-1}/ 22500 MW = 45%**

(e.g.) Agua Caliente Solar Project (AZ, US) has an installed capacity of 290 MW and an average annual production of 740 GWh yr^{-1}. Its capacity factor is **= %**

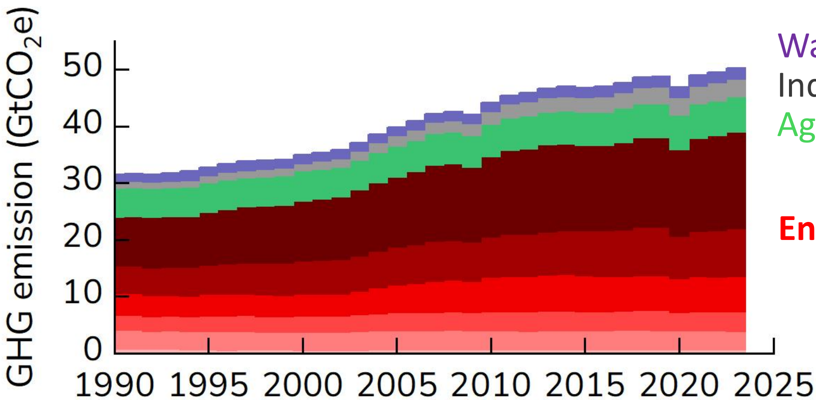
	Solar	Wind (Onshore)	(Offshore)	Hydropower	Geothermal	Biomass	Nuclear
Installed Cost (\$ kW^{-1})	691	1,041	2,852	2,267	4,015	3,242	~5,500
LCOE (\$ MWh^{-1})	43	34	79	57	60	87	~75
CF (%)	17	34	42	48	88	73	~90

<Source: IRENA–Renewable Power Generation Costs in 2024 & IAEA>

Why Renewables?

❑ Global GHG Emissions by Sector

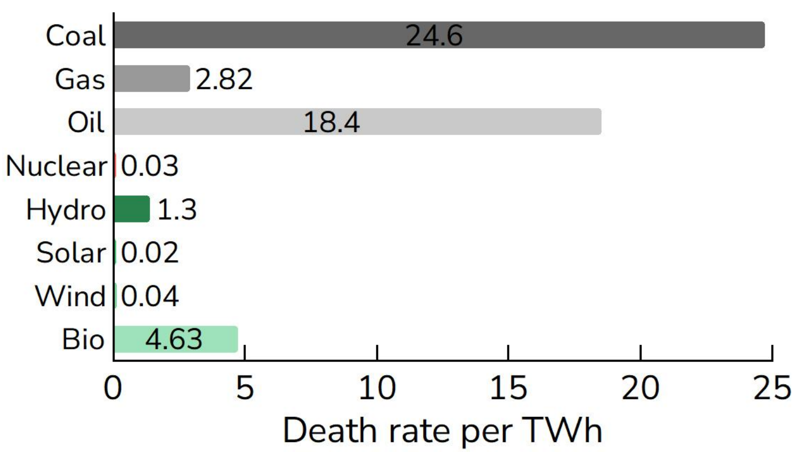
Greenhouse gas emissions in 2023 were ~50 billion tonnes CO_2eq, with ~80% coming from *energy*



Waste (3.9%) (Wastewater, Landfill, etc.)
Industry (6.3%) (e.g., Cement, Chemical, etc.)
Agriculture, Forestry (12%) (e.g., Livestock, Deforestation, etc.)

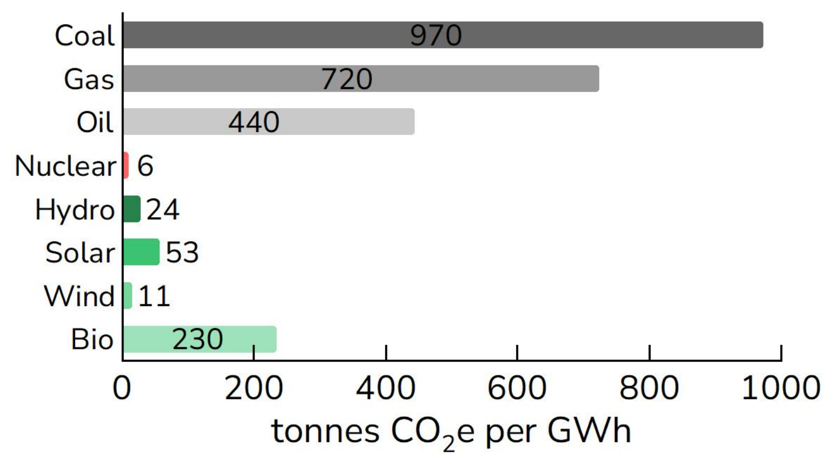
Energy (77%) (e.g., Electricity, Heat, Transport, etc.)

❑ Air Pollution & Accident



<Source: Our World in Data>

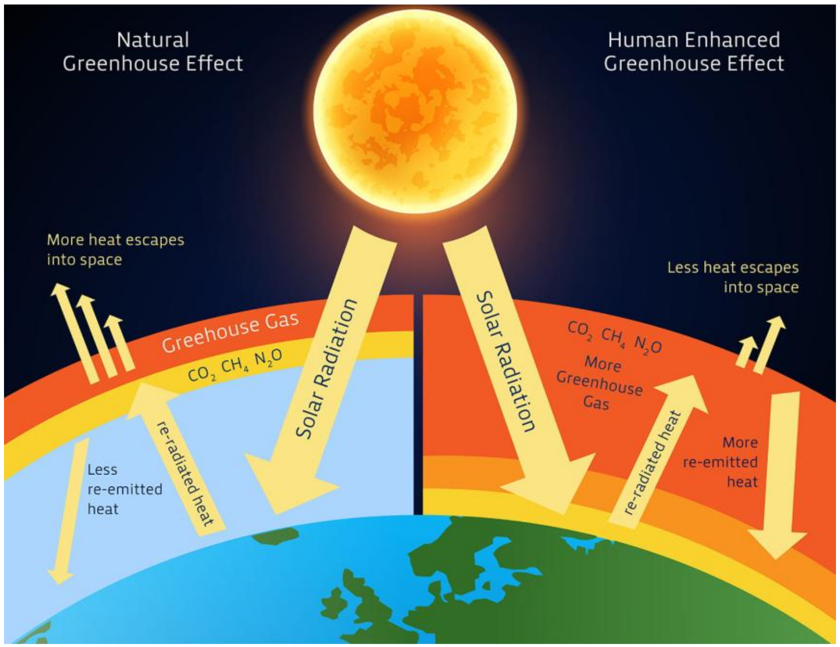
❑ GHG Emissions



Greenhouse Effect

❑ Greenhouse Effect

GHGs prevent the planet from losing heat to space



Renew. Sustain. Energy Rev., **71** (2017) 112–126

❑ Earth's Energy Budget

Annual Earth Energy Imbalance = ~0.6 W m^{-2}

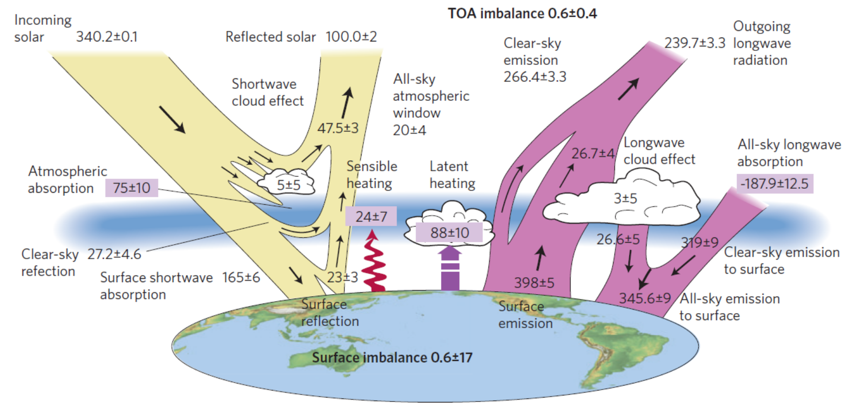


Fig. Global annual mean energy budget of Earth for the approximate period 2000–2010.

Yellow: Solar fluxes
Pink: Infrared fluxes

Nat. Geosci., **5** (2012) 691–696

– The surface of the Sun is ~5500 °C, so it emits *shortwave* radiation (UV, VL, IR)

– Shortwave radiation passes through GHGs and heats the Earth's surface

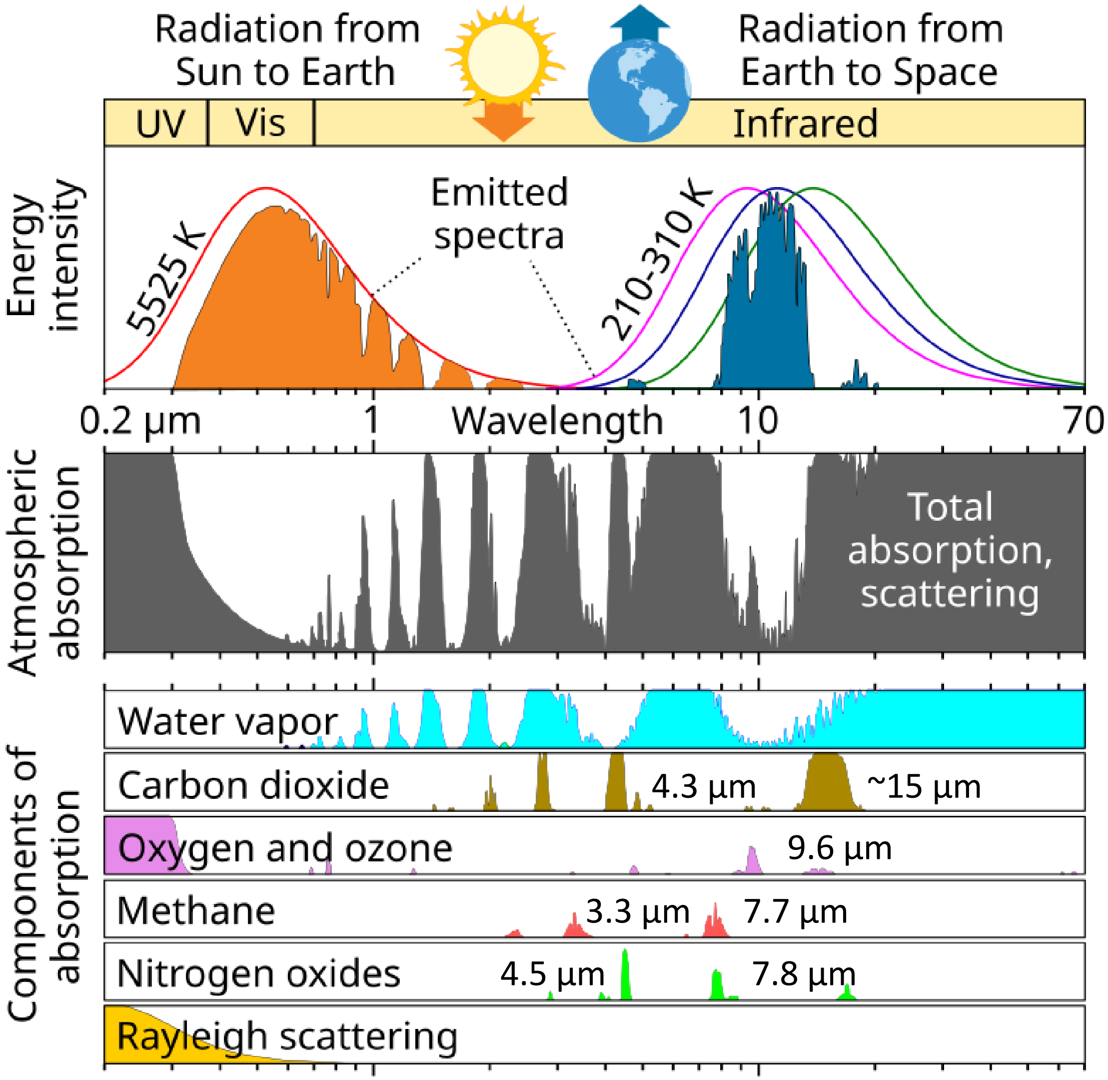
– The Earth's surface emits *longwave* radiation that is mostly absorbed by GHGs

Greenhouse Effect

❑ Atmosphere of Earth

Gas	Volume fraction (ppm)
N_2	780,840
O_2	209,460
Ar	9,340
CO_2	**420**
Ne	18.182
He	5.24
CH_4	**1.92**
Kr	1.14
H_2	0.55
N_2O	**0.33**
CO	**0.10**
Xe	0.09
O_3	**0.07**
NO_2	**0.02**
H_2O	**Up to 40,000**

<Source: National Oceanic and Atmospheric Administration>

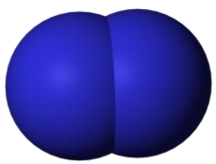


Greenhouse Effect

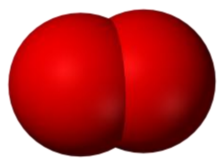
❑ Infrared Inactive Molecules

Gases with single or two identical atoms

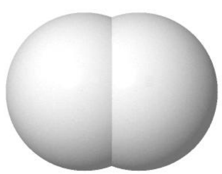
Nitrogen (N_2)



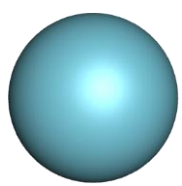
Oxygen (O_2)



Hydrogen (H_2)



Helium (He)



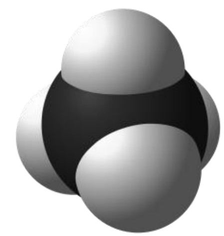
– Transparent to longwave radiation

– No net change in dipole moment

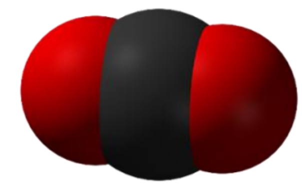
❑ Infrared Active Molecules

Gases with two or more different atoms

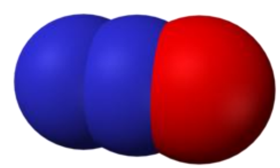
Methane (CH_4)



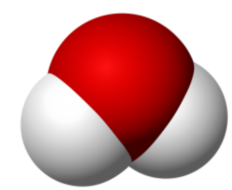
Carbon dioxide (CO_2)



Nitrous oxide (N_2O)



Water/Steam (H_2O)



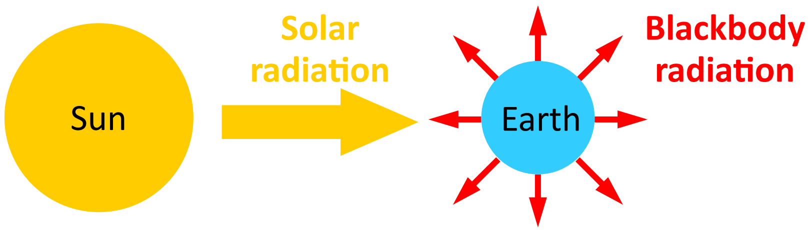
– Absorbing IR radiation in the bonds

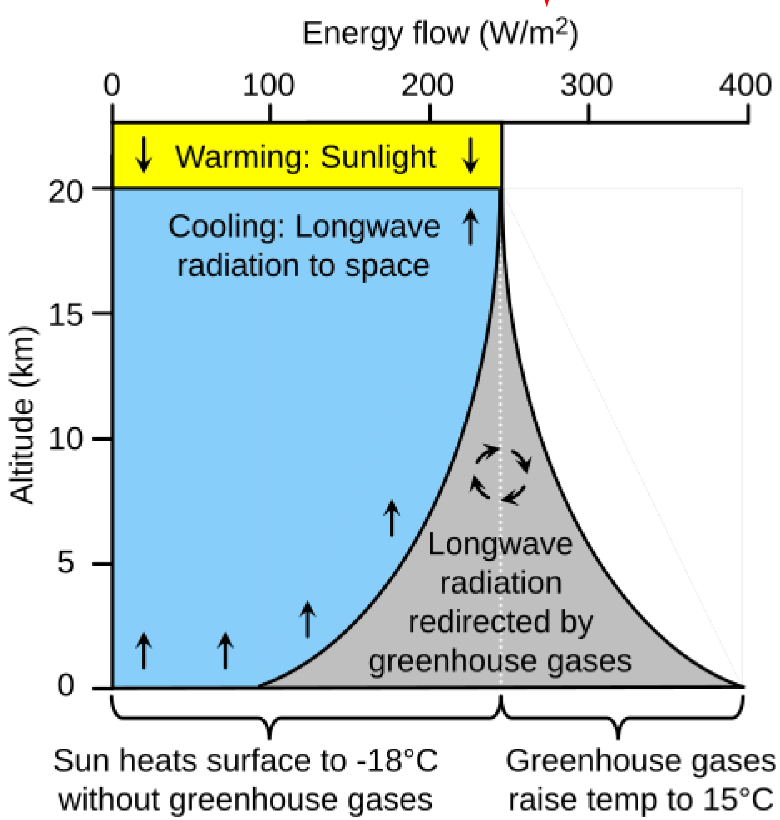
– Vibrations modify the dipole moment

Greenhouse Effect

❑ What if…

Without GHGs, the average surface temperature of the Earth would be as cold as ~–19 °C





Under the black-body approximation, the Sun's luminosity $L_\odot$ is

$$L_\odot = 4\pi R_\odot^2 \sigma T_\odot^4 \quad (1)$$

, where $R_\odot$ and $T_\odot$ are respectively the radius and temperature of the Sun and σ is the Stefan–Boltzmann constant.

This energy is passing through a sphere with a radius of a_0, the distance between the Earth and the Sun (i.e., 1 AU), and the irradiance (received power per unit area) $E_\oplus$ is

$$E_\oplus = \frac{L_\odot}{4\pi a_0^2} \quad (2)$$

The radiant flux (i.e. solar power) absorbed by the Earth is

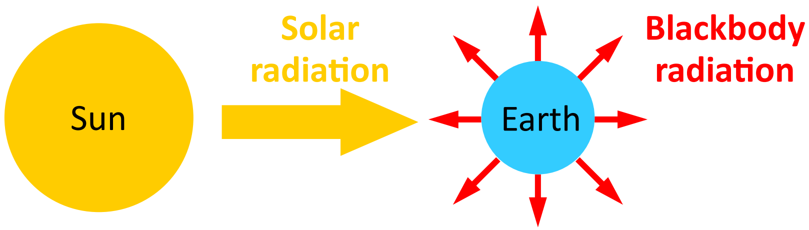
$$\Phi_{\rm abs} = 4\pi R_\oplus^2 Q_0 = \pi R_\oplus^2 E_\oplus \quad (3)$$

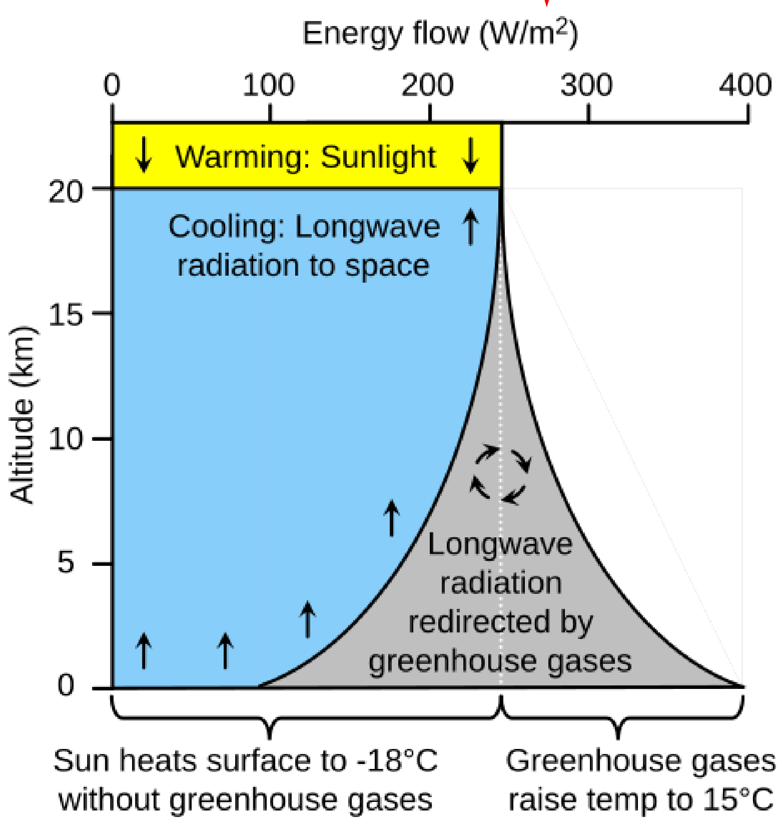
, where $R_\oplus$ is the radius of the Earth and Q_0 is global mean solar irradiance.

Greenhouse Effect

❑ What if…

Without GHGs, the average surface temperature of the Earth would be as cold as ~–19 °C





Because the Stefan–Boltzmann law uses a fourth power, it has a stabilizing effect on the exchange, and the flux emitted by Earth tends to be equal to the flux absorbed, close to the steady state

$$4\pi R_\oplus^2 \sigma T_\oplus^4 = \pi R_\oplus^2 E_\oplus \qquad (4)$$

$$= \pi R_\oplus^2 \frac{4\pi R_\odot^2 \sigma T_\odot^4}{4\pi a_0^2} \qquad (5)$$

The effective temperature of the Earth $T_\oplus$ can be expressed by

$$T_\oplus^4 = \frac{R_\odot^2 T_\odot^4}{4a_0^2} \Leftrightarrow T_\oplus = T_\odot \sqrt{\frac{R_\odot}{2a_0}} \qquad (6)$$

Since $R_\odot$ = 6.957 × 10^8 m, $T_\odot$ = 5780 K, and a_0 = 1.496 × 10^{11} m, $T_\oplus$ is estimated as to be 278.7 K (=5.564 °C), assuming that it perfectly absorbs all emission falling on it and has no atmosphere.

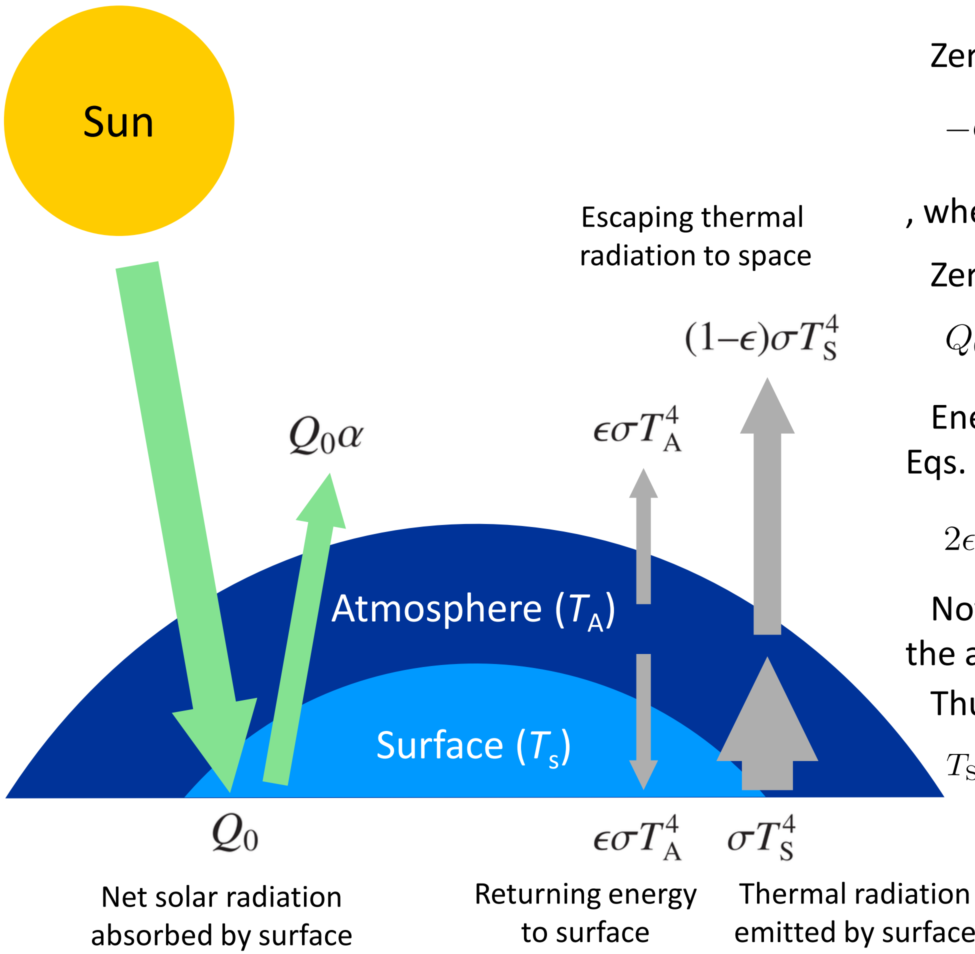
The Earth has an albedo α of 0.306, reducing the temperature by a factor of $0.694^{1/4}$, giving **254.4 K (=–18.76 °C)**.

Because of the greenhouse effect, the Earth's actual average surface temperature is **~287.2 K (~14.1 °C)**, which is higher than the estimated effective temperature, and even higher than the 297 K temperature that a blackbody would have.

Greenhouse Effect

Idealized Greenhouse Model

A simple model for mechanistically understanding the greenhouse effect on Earth



https://mintapps.org/html/mint-radiation-balance.html

Zero net radiation leaving the top of the atmosphere requires

$$-Q_0(1-\alpha)+\epsilon\sigma T_A^4+(1-\epsilon)\sigma T_S^4=0 \qquad (7)$$

, where ε is the emissivity of the atmosphere.

Zero net radiation entering the surface requires

$$Q_0(1-\alpha)+\epsilon\sigma T_A^4-\sigma T_S^4=0 \qquad (8)$$

Energy equilibrium of the atmosphere can be derived from Eqs. (7) and (8),

$$2\epsilon\sigma T_A^4-\epsilon\sigma T_S^4=0 \qquad (9)$$

Note the important factor of 2, resulting from the fact that the atmosphere radiates both upward and downward.

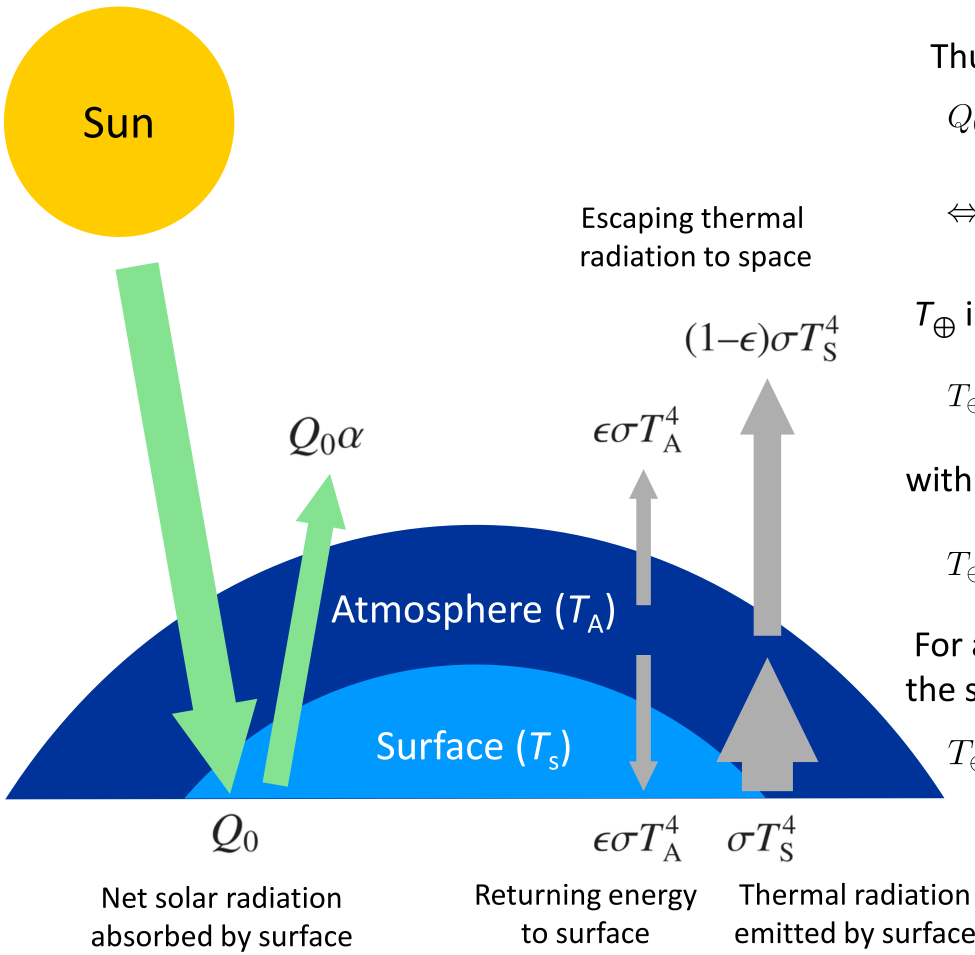
Thus, the ratio of T_A to T_S is independent of ε

$$T_S=2^{0.25}T_A=1.189T_A \qquad (10)$$

Greenhouse Effect

❑ Idealized Greenhouse Model

A simple model for mechanistically understanding the greenhouse effect on Earth



https://mintapps.org/html/mint-radiation-balance.html

Thus, T_A can be expressed in terms of T_S, and then Eq. (8) gives

$$Q_0(1-\alpha) = (1-\frac{\epsilon}{2})\sigma T_S^4 \quad (11)$$

$$\Leftrightarrow T_S = \left[\frac{Q_0(1-\alpha)}{\sigma}\frac{1}{1-\frac{\epsilon}{2}}\right]^{\frac{1}{4}} \quad (12)$$

$T_\oplus$ is the solution for T_S, for the case of $\varepsilon = 0$ (no atmosphere)

$$T_\oplus = \left[\frac{Q_0(1-\alpha)}{\sigma}\right]^{\frac{1}{4}} \quad (13)$$

with the definition of $T_\oplus$:

$$T_\oplus = T_S\left[\frac{1}{1-\frac{\epsilon}{2}}\right]^{\frac{1}{4}} \quad (14)$$

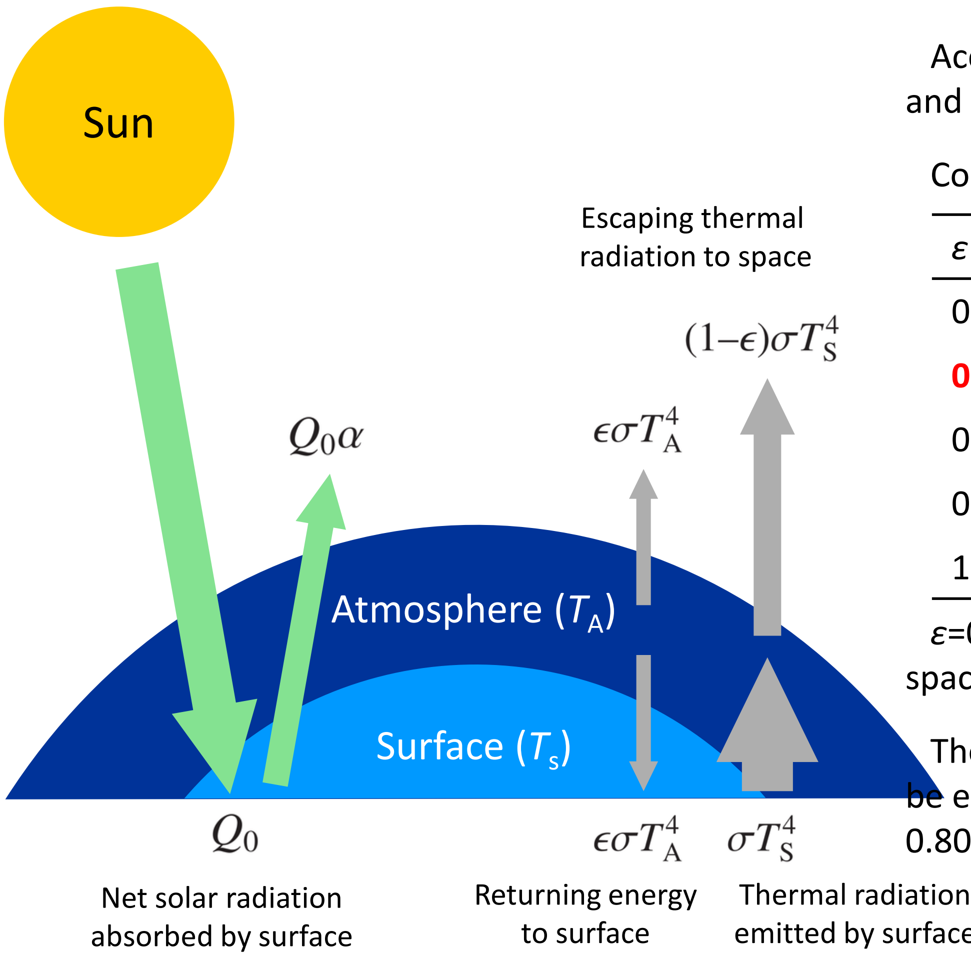
For a perfect greenhouse, with no radiation escaping from the surface, or ε=1:

$$T_\oplus = T_A \quad (15)$$

Greenhouse Effect

❑ Idealized Greenhouse Model

A simple model for mechanistically understanding the greenhouse effect on Earth



https://mintapps.org/html/mint-radiation-balance.html

According to Eqs. (1) and (2), $E_\oplus$ is estimated as 1369 W m^{-2}, and thus Q_0 is 342.2 W m^{-2} (σ = 5.67×10^{-8} W m^{-2} K^{-4})

Considering α is 0.306, T_S can be calculated as below (Eq. (12)):

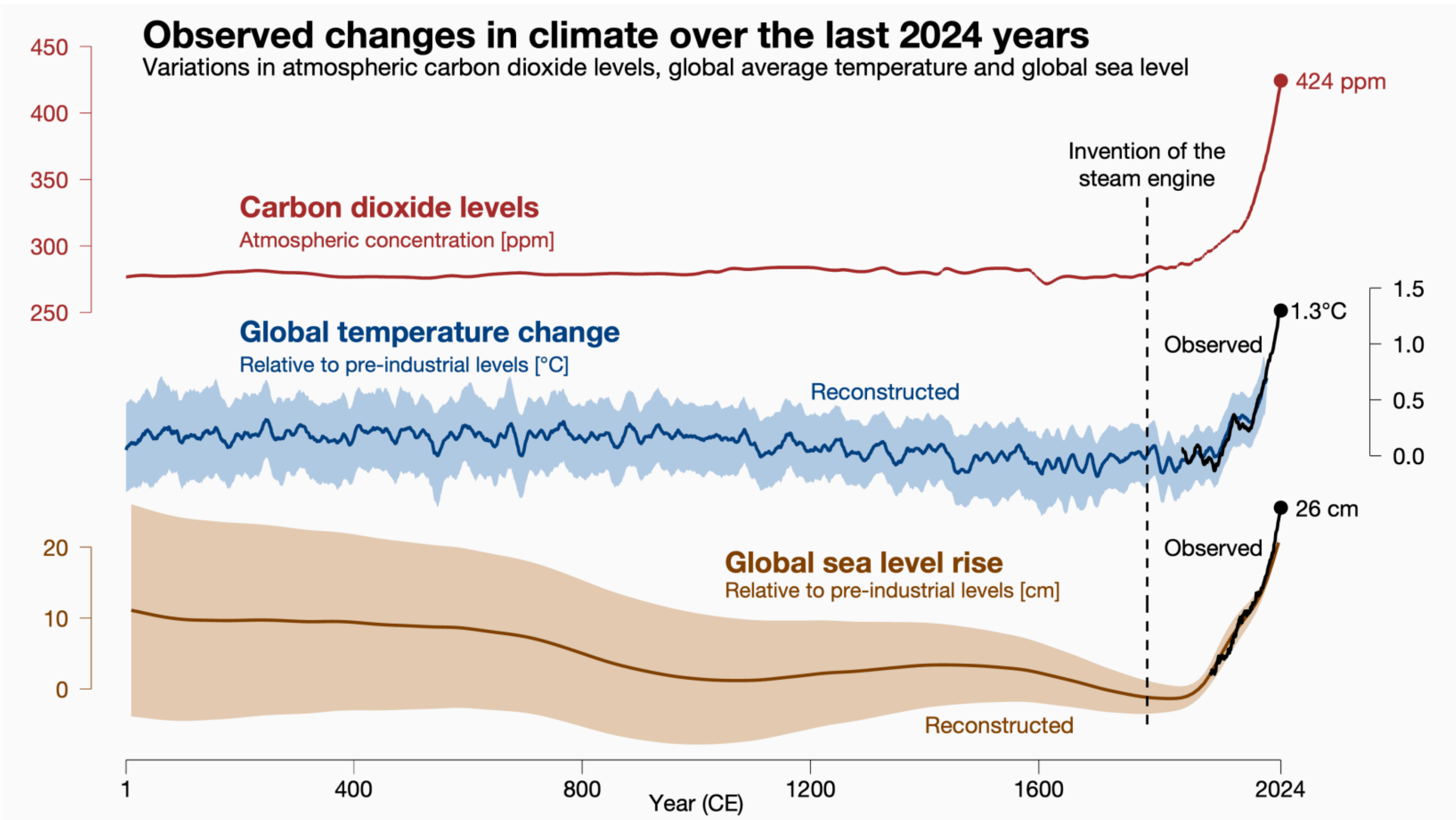
ε (–)	T_S (K)	T_S (°C)
0	254.4	–18.76
0.78	**287.8**	**14.70**
0.80	289.0	15.89
0.82	290.3	17.11
1	302.5	29.37

ε=0.78 implies 22% of the surface radiation escapes directly to space.

The radiative forcing for doubling CO_2 is ~3.71 W m^{-2}, which can be estimated by using $\Delta\varepsilon$ = 0.019. Thus, a change of ε from 0.78 to 0.80 is consistent with the radiative forcing from a doubling of CO_2.

Greenhouse Effect

❑ Atmospheric CO_2 and Global Temperature

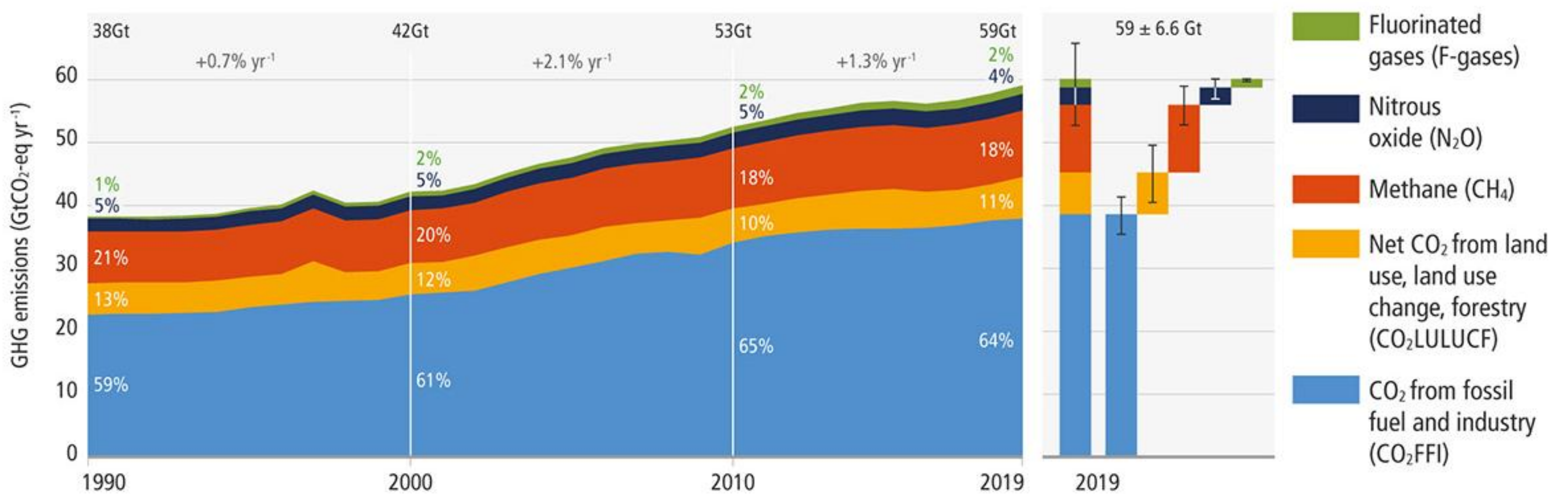


<Source: Climate Visuals>

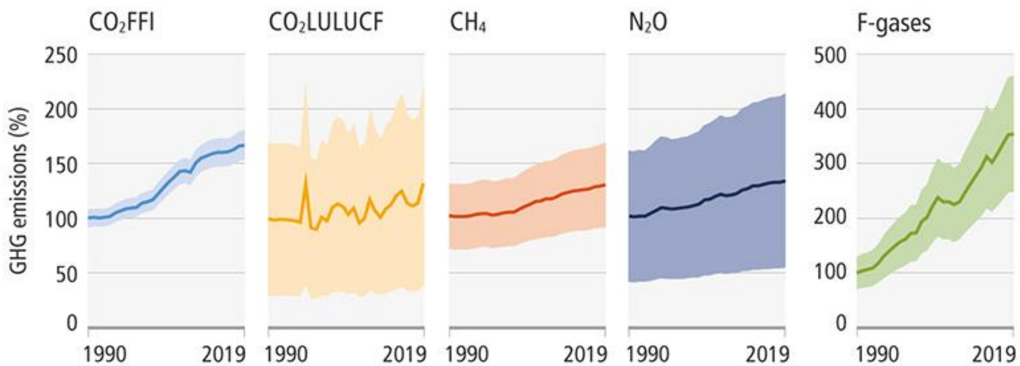
Greenhouse Effect

❑ Global Net Anthropogenic GHG Emissions

a. Global net anthropogenic GHG emissions 1990–2019 [5]



b. Global anthropogenic GHG emissions and uncertainties by gas – relative to 1990



	2019 emissions ($GtCO_2$-eq)	1990–2019 increase ($GtCO_2$-eq)	Emissions in 2019, relative to 1990 (%)
CO_2 FFI	38±3	15	167
CO_2 LULUCF	6.6±4.6	1.6	133
CH_4	11±3.2	2.4	129
N_2O	2.7±1.6	0.65	133
F-gases	1.4±0.41	0.97	354
Total	59±6.6	21	154

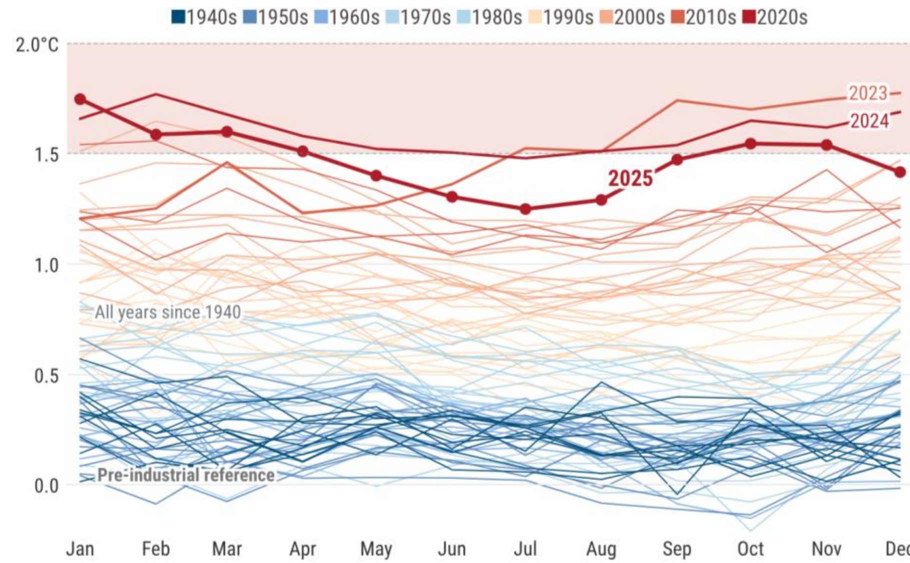
The solid line indicates central estimate of emissions trends. The shaded area indicates the uncertainty range.

<Source: IPCC>

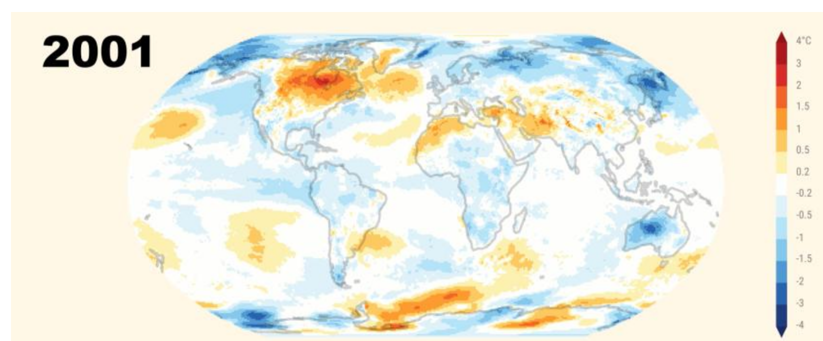
Climate Change

❏ Global Boiling

Monthly global surface air temperature anomalies



Annual surface air temperature anomalies (2001–2025)



<Source: Corpernicus Climate Change (2025)>

Climate Change-Driven Disasters

- ❑ **Heatwave**

Hajj extreme heat disaster (Jun 2024)

- ❑ **Tropical Cyclone**

Hurricane Helene (Sep 2024)

- ❑ **Wildfire**

LA wildfire (Jan 2025)

- ❑ **Drought**

Amazon deforestation

- ❑ **Flood**

Spanish floods (Oct–Nov 2024)

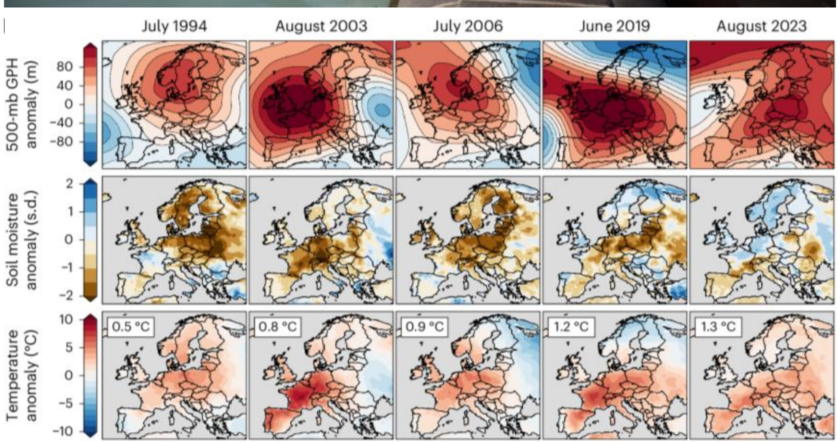
- ❑ **Particulate matter**

Air pollution in London

Climate Change-Driven Disasters

❑ European Heatwaves

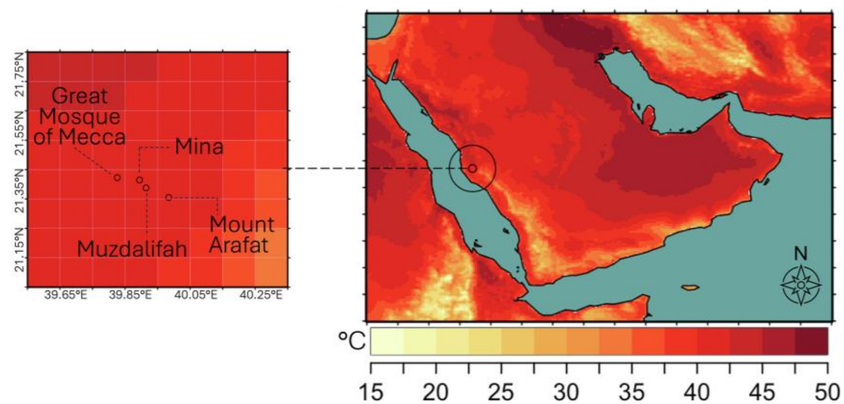
More than 62,000 people died in Europe's 2024 heatwave: Which country was hit the hardest?



Nat. Clim. Change, **16** (2026) 26–32

❑ Hajj Extreme Heat Disaster

At least 1,301 people died during Hajj - Saudi Arabia

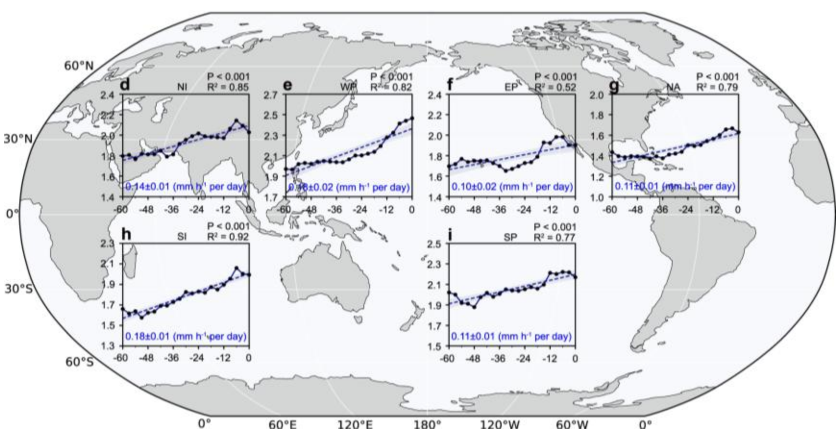


npj Nat. Hazards, **2** (2025) 107

Climate Change-Driven Disasters

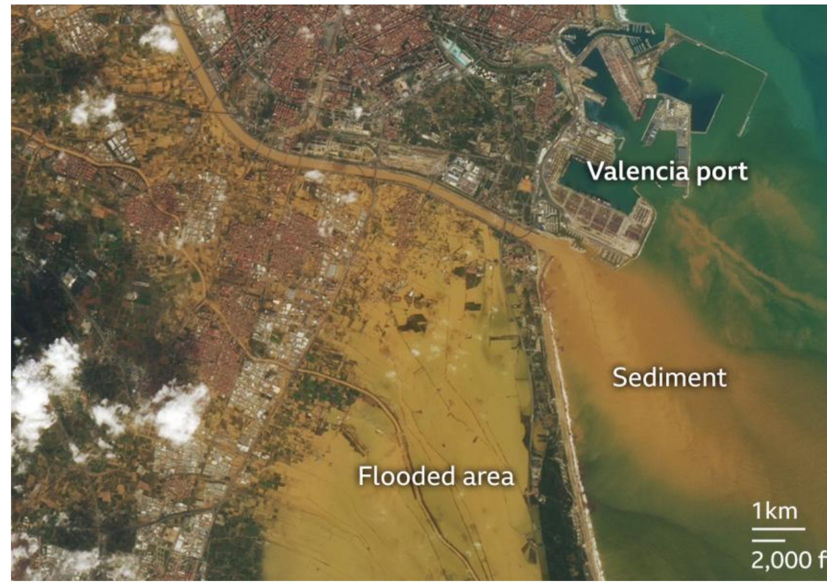
❑ Tropical Cyclone

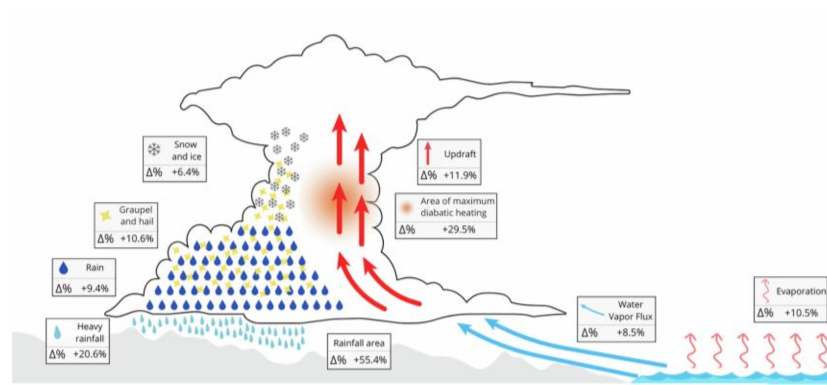
Ferocity of Atlantic hurricanes surges as the ocean warms



Nat. Comm., **17** (2026) 114

❑ Heavy Rainfalls



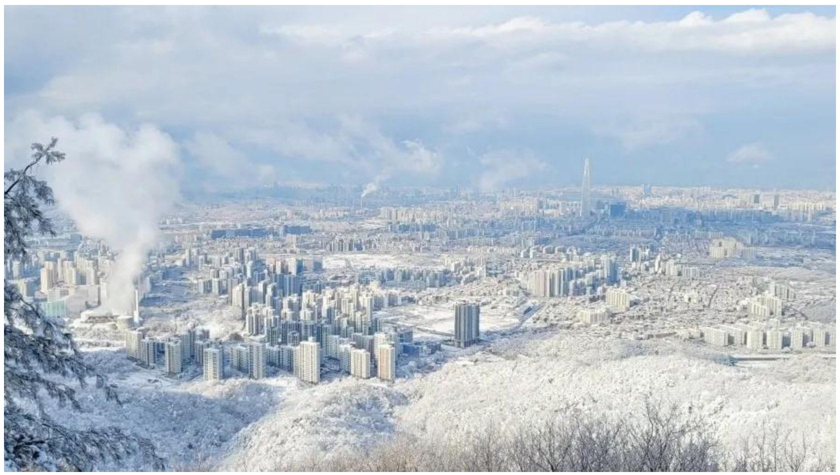


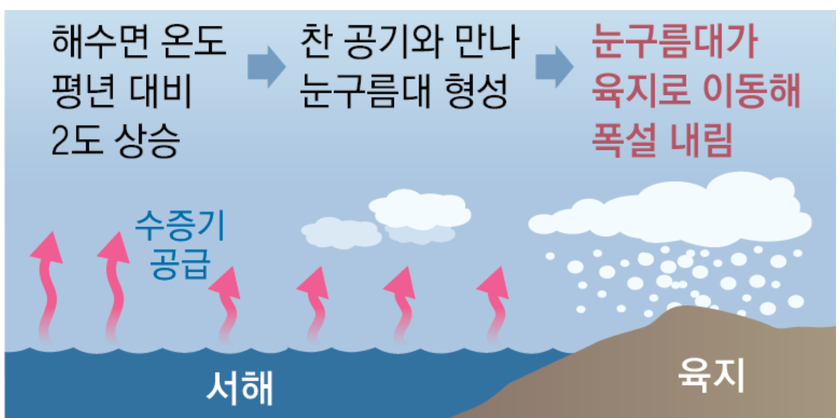
Nat. Comm., **17** (2026) 1492

Climate Change-Driven Disasters

❑ Snowstorm

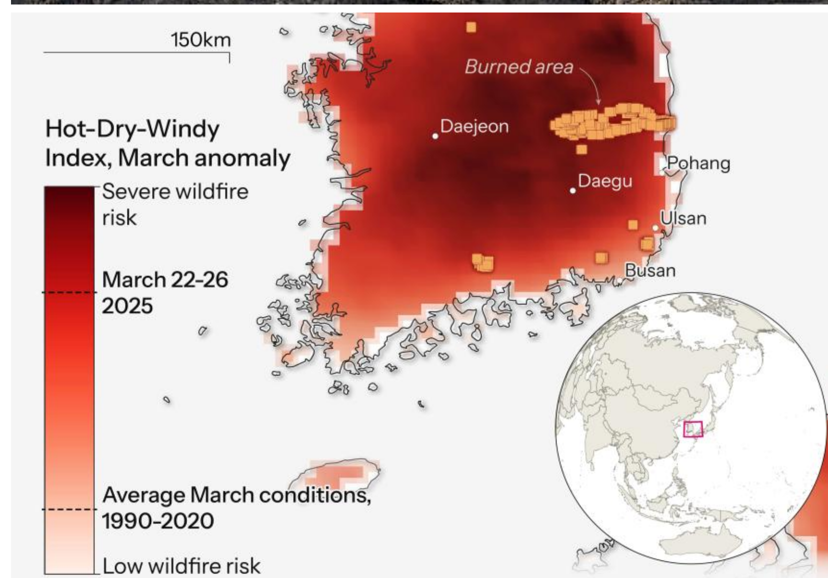
117년 만 11월 폭설, 서울 20cm 넘는 눈 처음





❑ Wildfire

영남 산불 피해면적 10만4천ha로 잠정집계



<Source: World Weather Attribution (WWA)>

Climate Impacts

❑ Temperature Rise Scenarios and Climate Change Impacts

	1.5°C	2°C	3°C	1.5°C vs 2°C	1.5°C vs 3°C
BIODIVERSITY LOSS Maximum percentage of species at high risk of extinction across forests and land	14%	18%	29%	1.3x WORSE	2.1x WORSE
DROUGHT Dryland population exposed to water stress, heat stress and desertification	0.95B PEOPLE	1.15B PEOPLE	1.29B PEOPLE	200M MORE PEOPLE	340M MORE PEOPLE
FOOD SECURITY Costs for adaptation and residual damage to major crops	$63 BILLION US	$80 BILLION US	$128 BILLION US	$17B MORE	$65B MORE
FIRES Increases in burnt area across Mediterranean Europe	40-54%	62-87%	96-187%	1.6x WORSE	3x WORSE
EXTREME HEAT Increase in number of days per year with a maximum temperature above 35°C (95°F)	45-58	52-68	66-87	1.2x WORSE	1.5x WORSE
EXTREME HEAT Increase in annual number of heatwaves in Southern Africa	2-4 TIMES	4-8 TIMES	8-12 TIMES	2x WORSE	3.3x WORSE

Climate Impacts

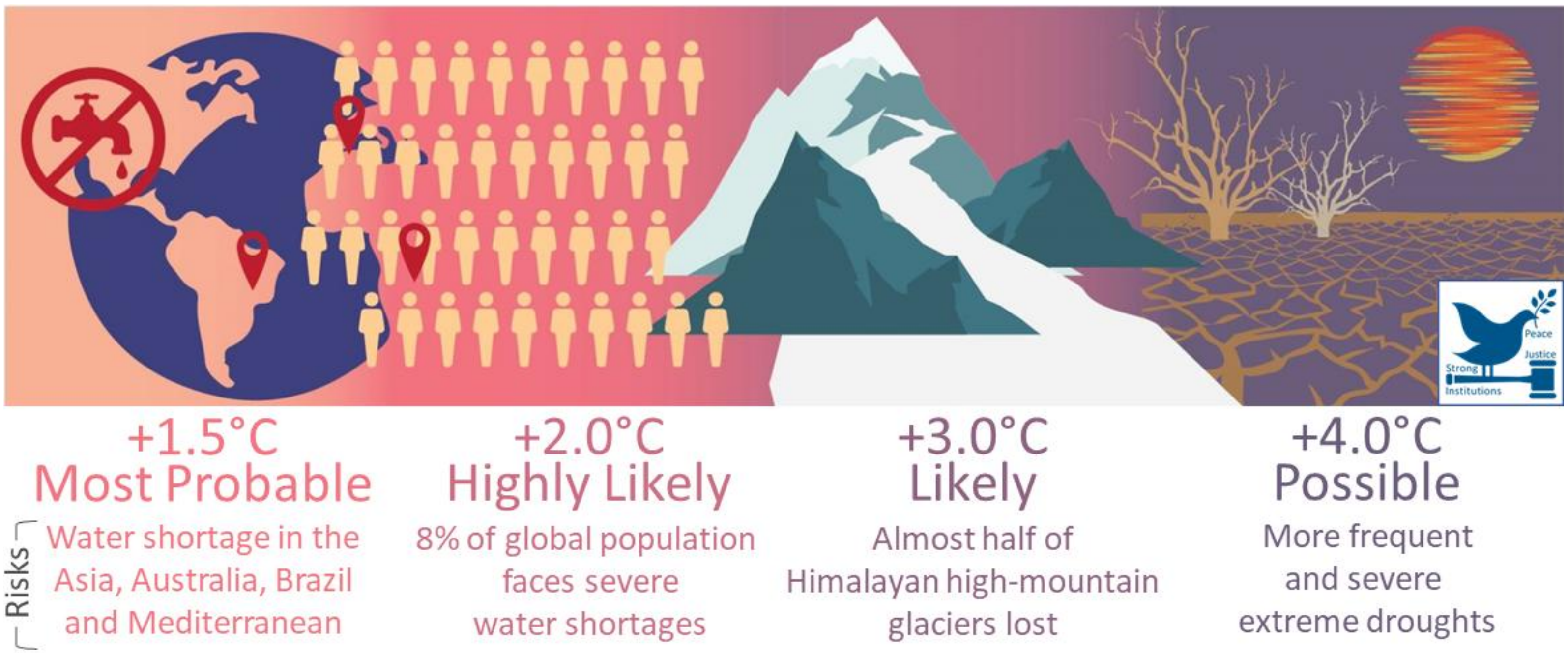
❑ Temperature Rise Scenarios and Climate Change Impacts (Cont'd)

	1.5°C	2°C	3°C	1.5°C vs 2°C	1.5°C vs 3°C
SEA LEVEL RISE Global mean sea level rise by 2100	0.28-0.55m	0.33-0.61m	0.44-0.76m	1.1x WORSE	1.4x WORSE
FLOODS Increase in global population exposed to flooding	24%	30%	NO DATA AVAILABLE	1.3x WORSE	NO DATA AVAILABLE
CORAL REEFS Further decline in coral reefs	70-90%	99%	NO DATA AVAILABLE	1.2x WORSE	NO DATA AVAILABLE

<Source: Climate Resources Institute>

Climate Impacts

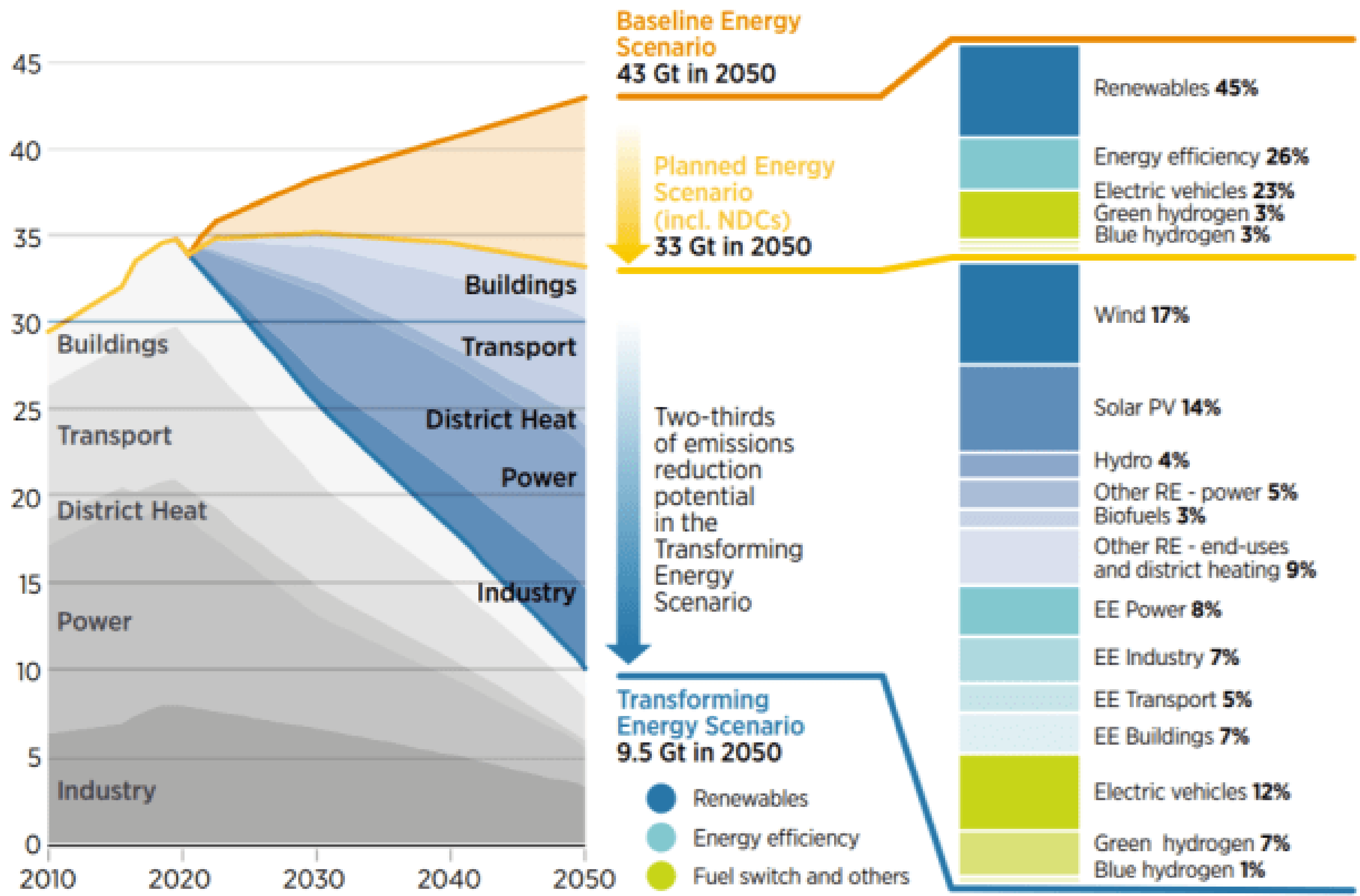
❑ Temperature Rise Scenarios and Climate Change Impacts



<Source: IPCC>

Pathways to 1.5 °C

❑ Mitigating GHG Emissions by Energy Transition



<Source: GLOBAL ENERGY TRANSFORMATION - a roadmap to 2050>

Assignment (~08th Sep)

❑ Planetary Equilibrium Temperature

Consider a planetary body orbiting the Sun at a distance a with an albedo α and radius R.
The Sun has a luminosity $L_{\odot}=4\pi R_{\odot}^2 \sigma T_{\odot}^4$, where $R_{\odot}$ is the radius and $T_{\odot}$ is the effective surface temperature of the Sun. Solve the following problems:

- Express the solar irradiance E (received power per unit area) at distance a, and write down the net solar radiation Q_0 entering the planetary surface using σ, $T_{\odot}$, $R_{\odot}$, and a.

- Derive the theoretical effective equilibrium temperature of the planet without an atmosphere T using $T_{\odot}$, $R_{\odot}$, a, and α.

Assignment (~08th Sep)

❑ Planetary Equilibrium Temperature (Cont'd)

Consider a planetary body orbiting the Sun at a distance a with an albedo α and radius R.
The Sun has a luminosity $L_\odot = 4\pi R_\odot^2 \sigma T_\odot^4$, where $R_\odot$ is the radius and $T_\odot$ is the effective surface temperature of the Sun. Solve the following problems:

- In the 1-layer idealized greenhouse model with atmospheric emissivity ε, derive the surface temperature T_S as a function of T and ε, and express the returning longwave radiation redirected to the surface ($\varepsilon\sigma T_A^4$, where T_A is the effective temperature of the atmosphere).

Assignment (~08th Sep)

❑ Planetary Equilibrium Temperature (Cont'd)

Consider a planetary body orbiting the Sun at a distance a with an albedo α and radius R.
The Sun has a luminosity $L_\odot = 4\pi R_\odot^2 \sigma T_\odot^4$, where $R_\odot$ is the radius and $T_\odot$ is the effective surface temperature of the Sun. Solve the following problems:

- Calculate the effective equilibrium temperature for Venus ($\alpha_V = 0.750$, $a_V = 0.723a_0$) T_V and compare its greenhouse effect based on its actual surface temperatures ($T_{S,V}$ = ~737 K).

Parameter	Value
$R_\odot$	6.96×10^8 m
$T_\odot$	5780 K
σ	5.67×10^{-8} W m^{-2} K^{-4}
A_0	1.50×10^{11} m

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